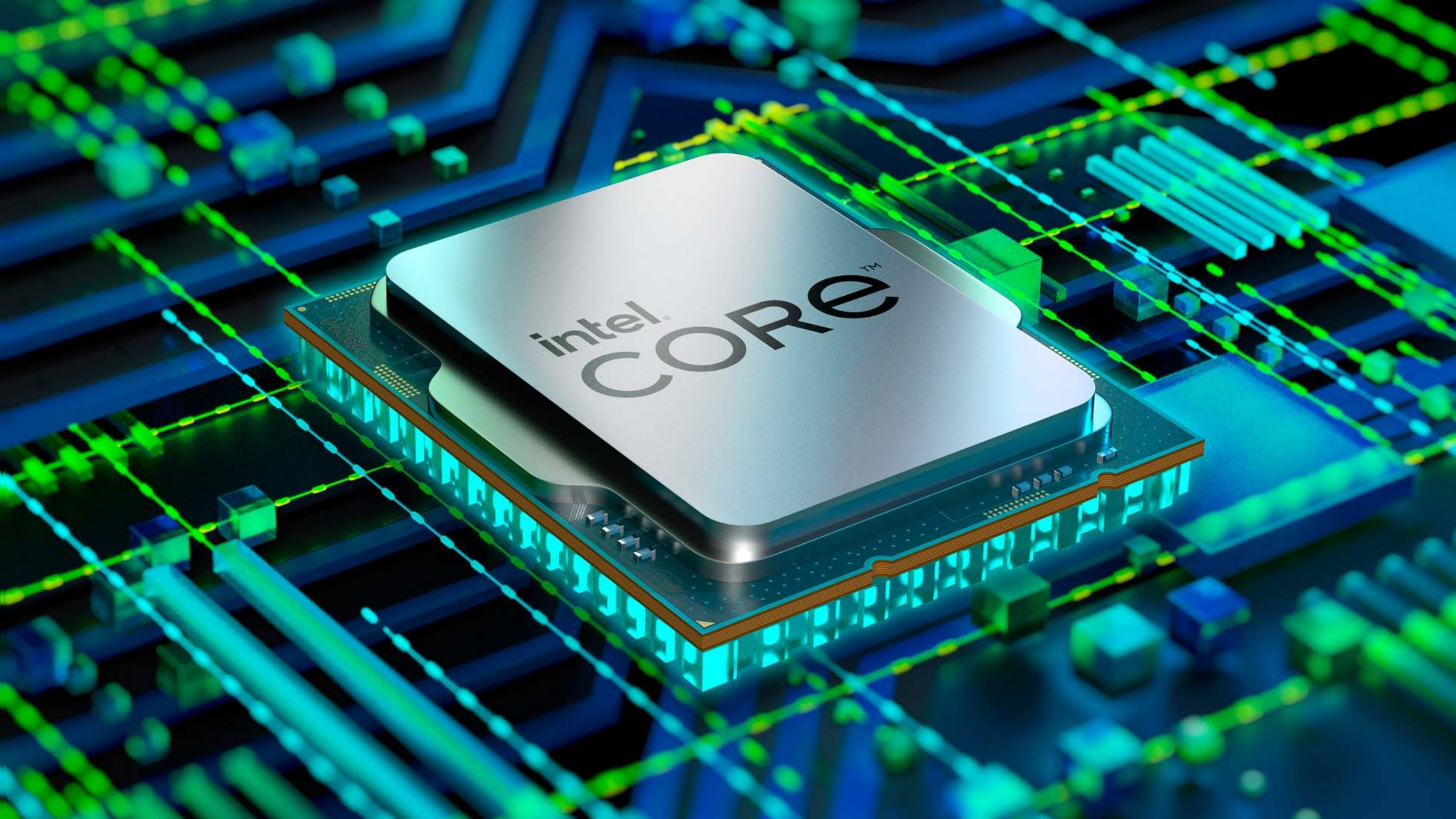




MICROCONTROLLER

AND ITS

APPLICATIONS

COURSE OBJECTIVES



- To explain the architecture of RP2040 chip
- To outline the features of cortex M0+ chip
- To explore the peripherals in RP2040



COURSE OUTCOME

- Describe the architecture of RP2040
- Illustrate the bus fabric and single cycle I/O of RP2040
- Use appropriate configurations of cortex M0+ in system design
- Demonstrate the Programmable I/O in system design.
- Categorize the peripherals for designing a system using RP2040



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Matlab - MIL/SII

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COURSE PLAN

Hands on!!!

CIAT 1

ASSIGN 1

CIAT 2

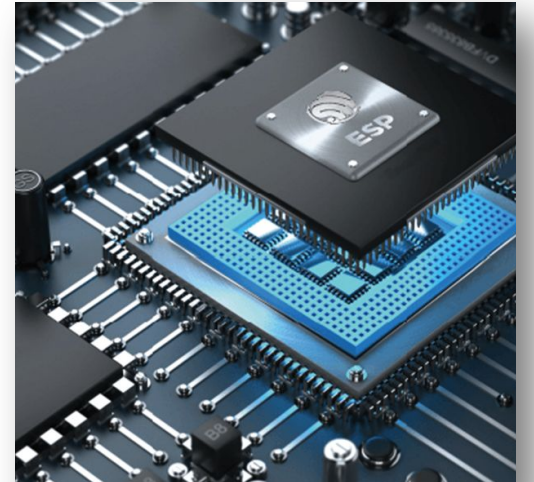
ASSIGN 2

CIAT 1 Unit 1,2 and Half in Unit 3

ASSIGN 1 Simulation and Study

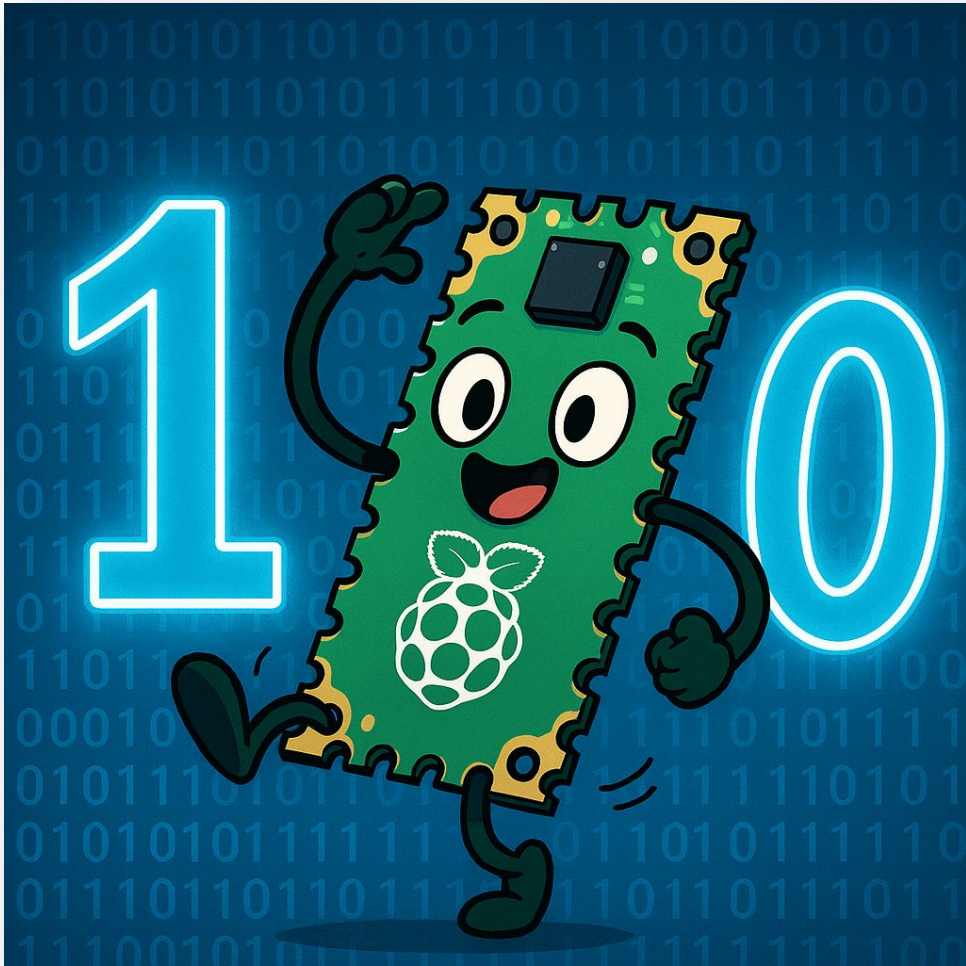
CIAT 2 Half in Unit 3, Unit 4 and 5

ASSIGN 2 Project and product



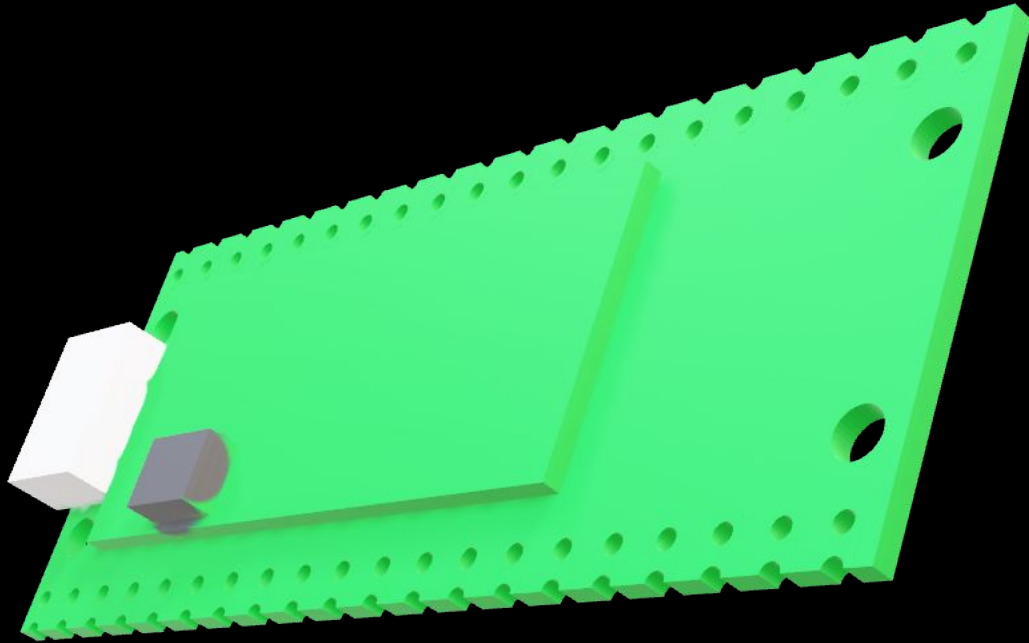
COURSE PLAN

Hands on!!!

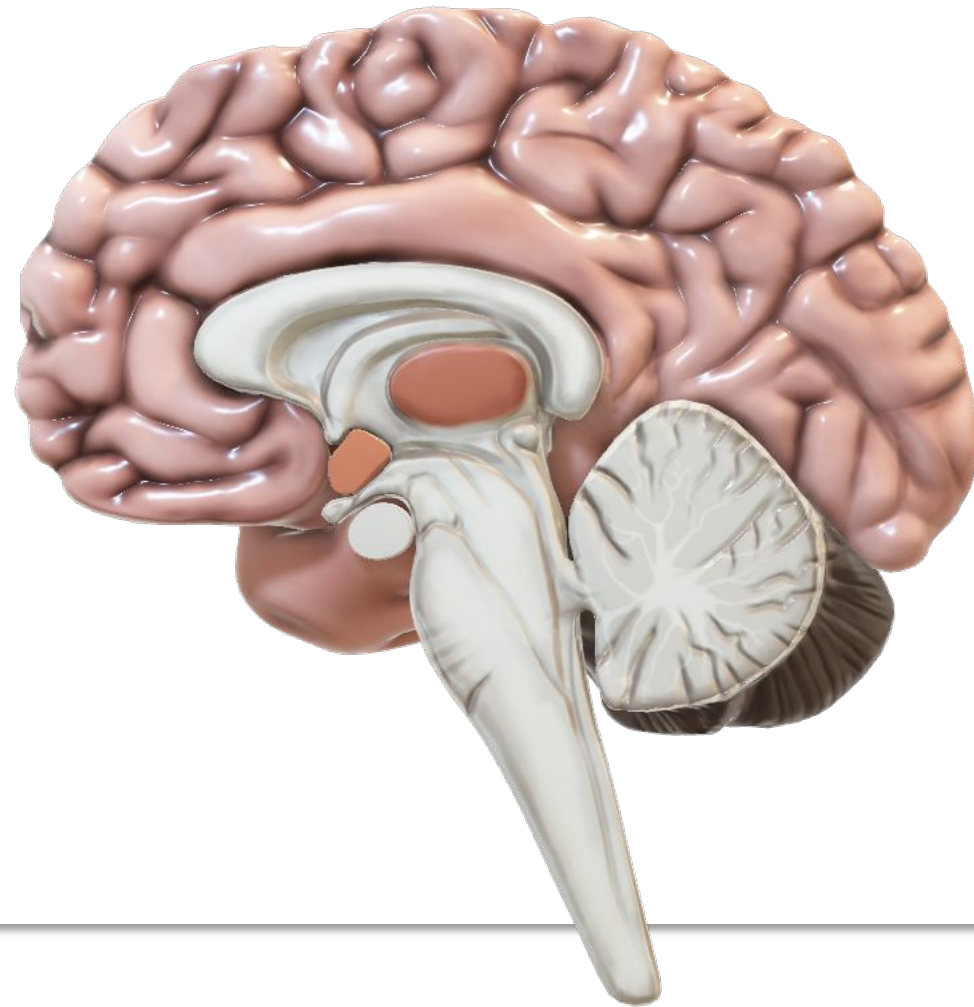


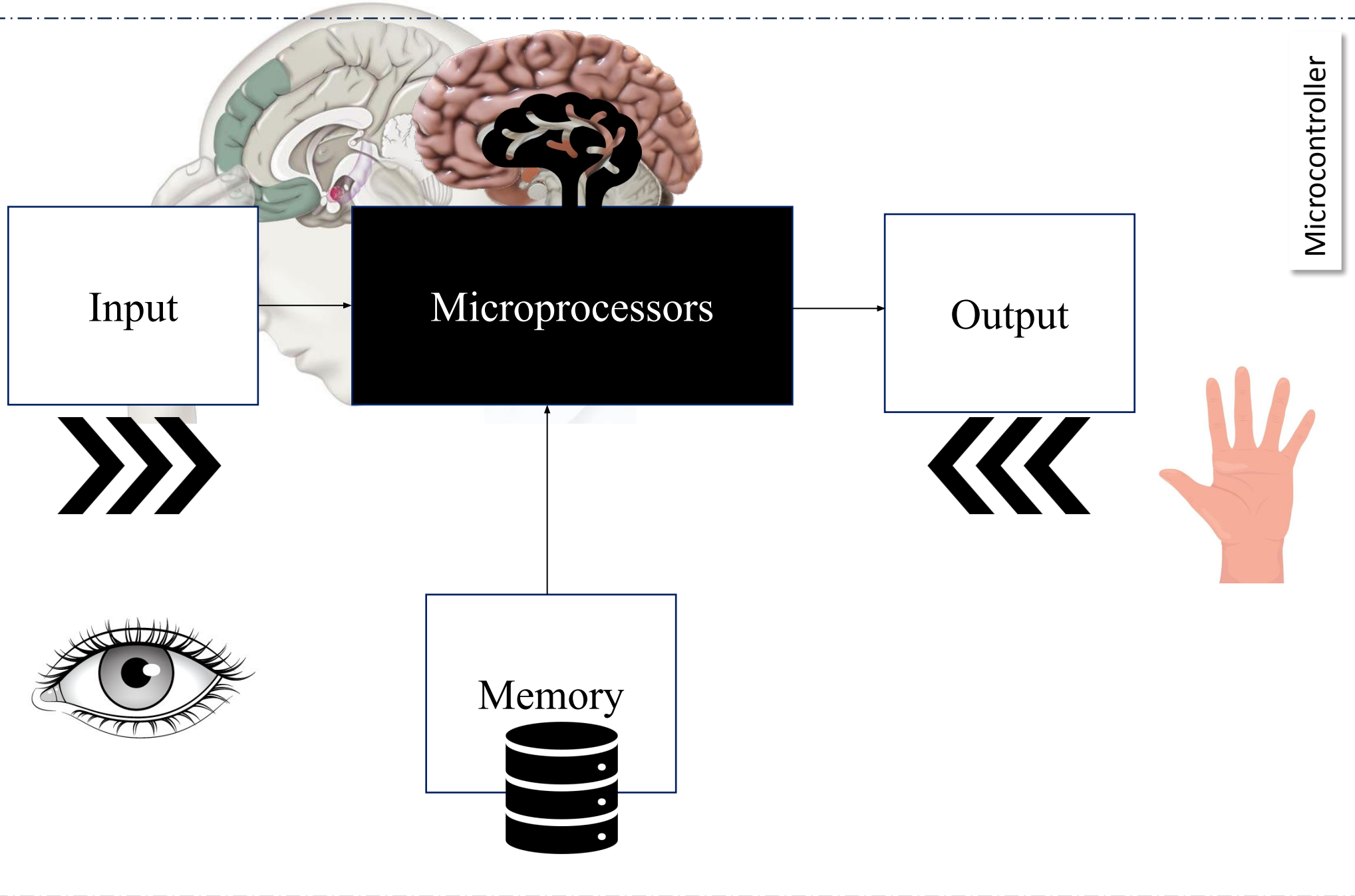
COURSE MATERIALS

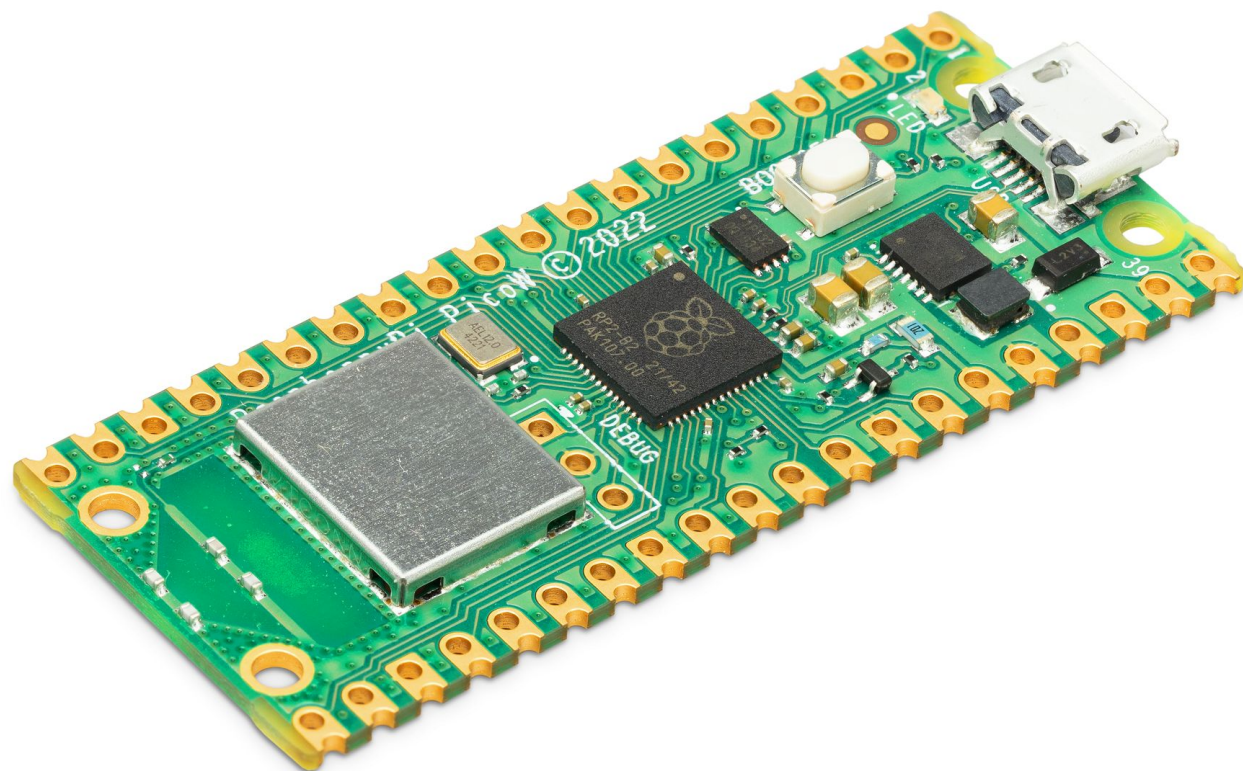




Raspberry Pi PICO



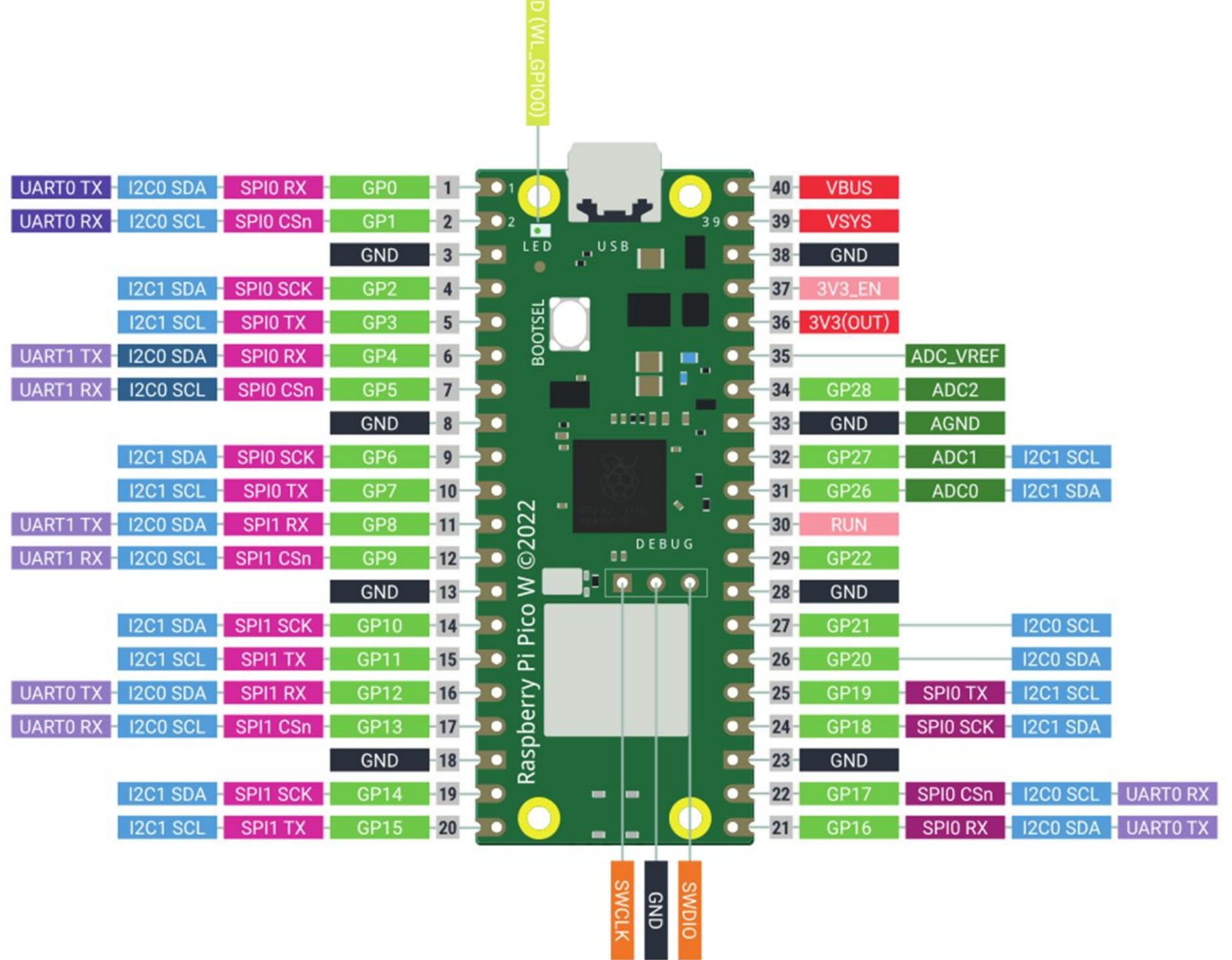




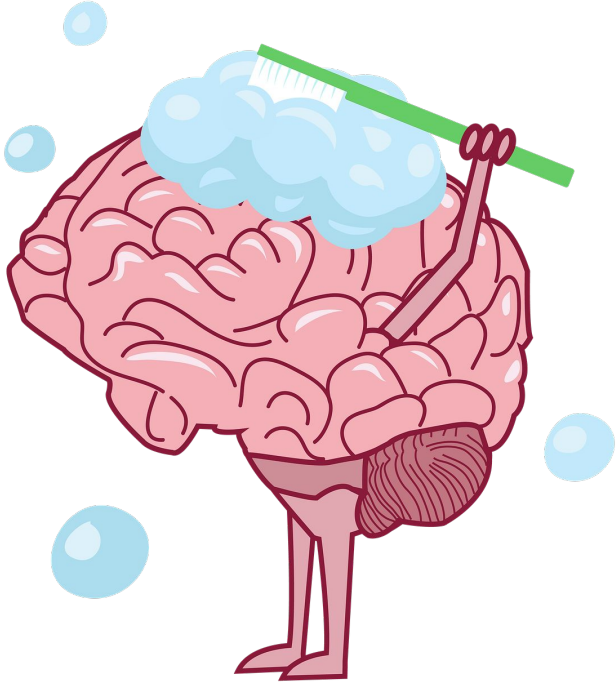
RP2040



Infineon 43439



Let's get our basics brushed



What is inside a microprocessor?

Millions of ICs?

What kinda ICs?

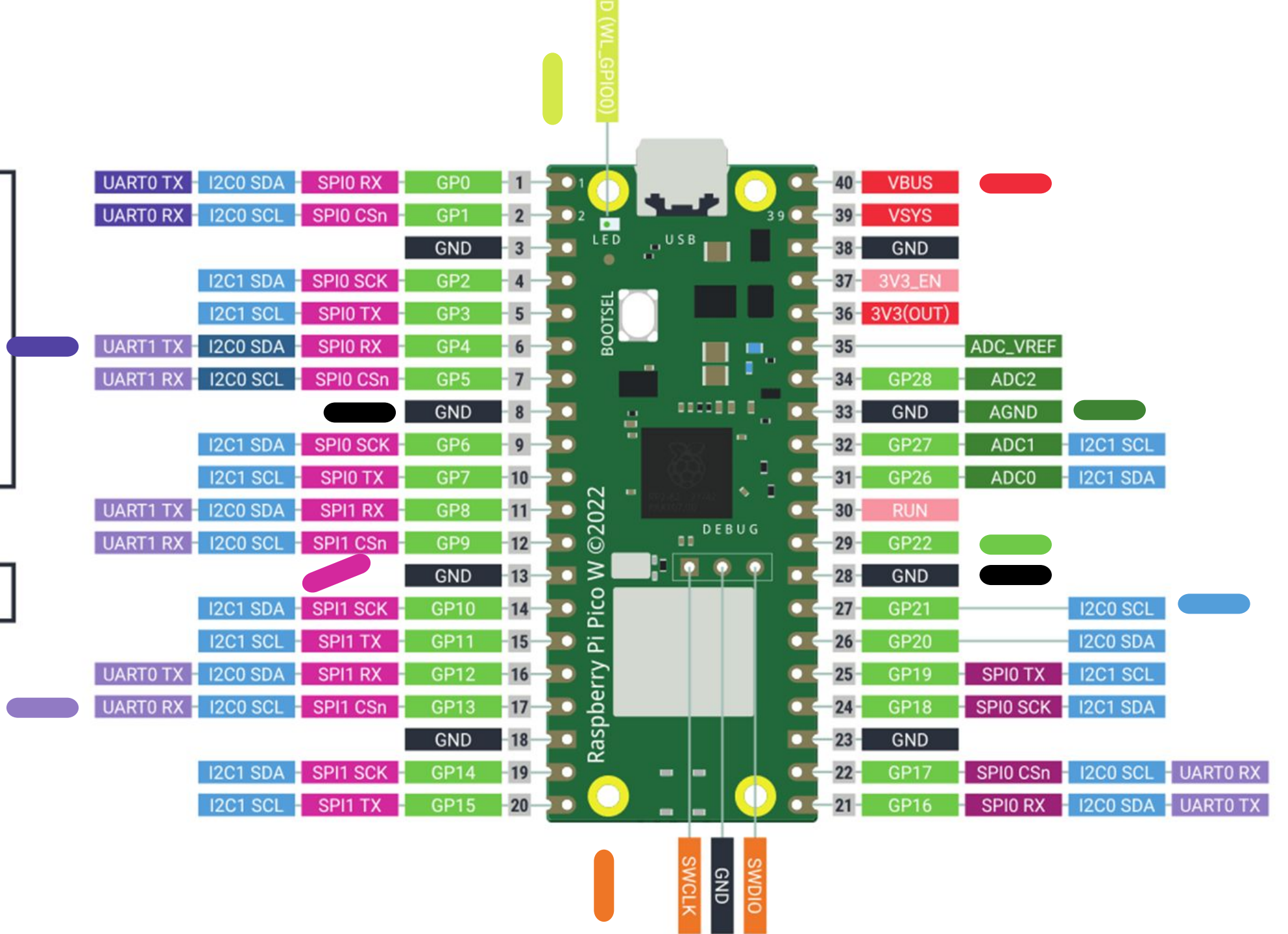
Analog IC | Digital IC

What is inside these ICs?

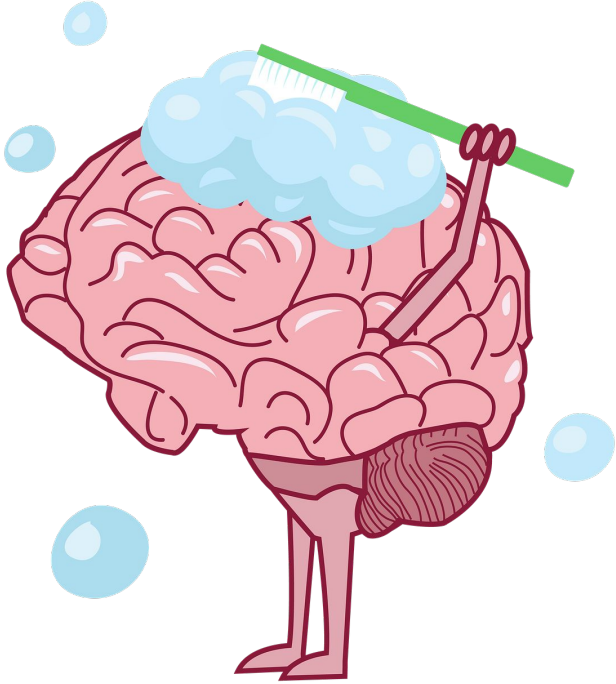
RP2040



Infinion 43439



Let's get our basics brushed



What is inside a microprocessor?

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Analog IC | Digital IC

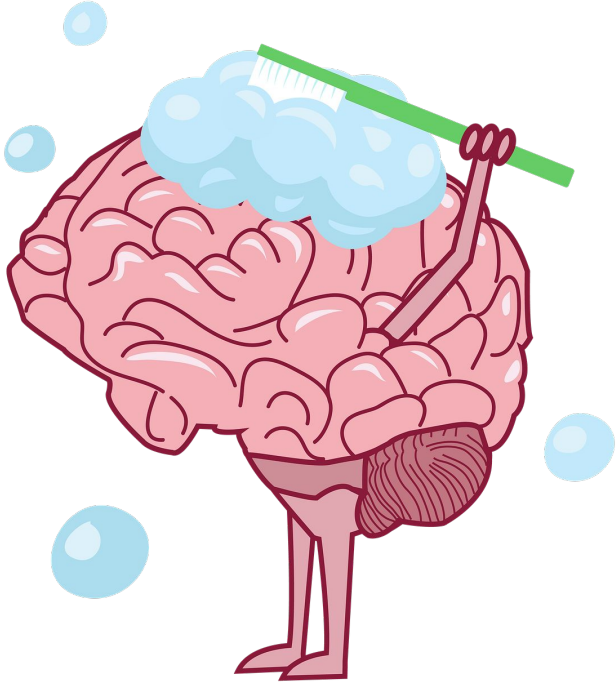
What is inside these ICs?

Transistors

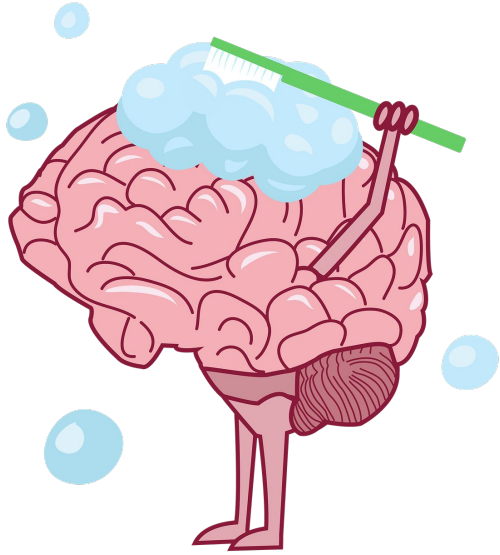
What is inside these transistors?

Diodes? FETs?

Let's get our basics brushed



Let's get our basics brushed



What is inside a microprocessor?

Millions of ICs?

What kinda ICs?

Analog IC | Digital IC

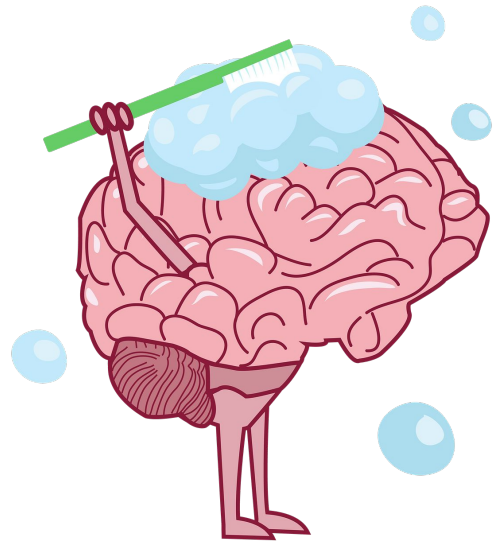
What is inside these ICs?

Transistors

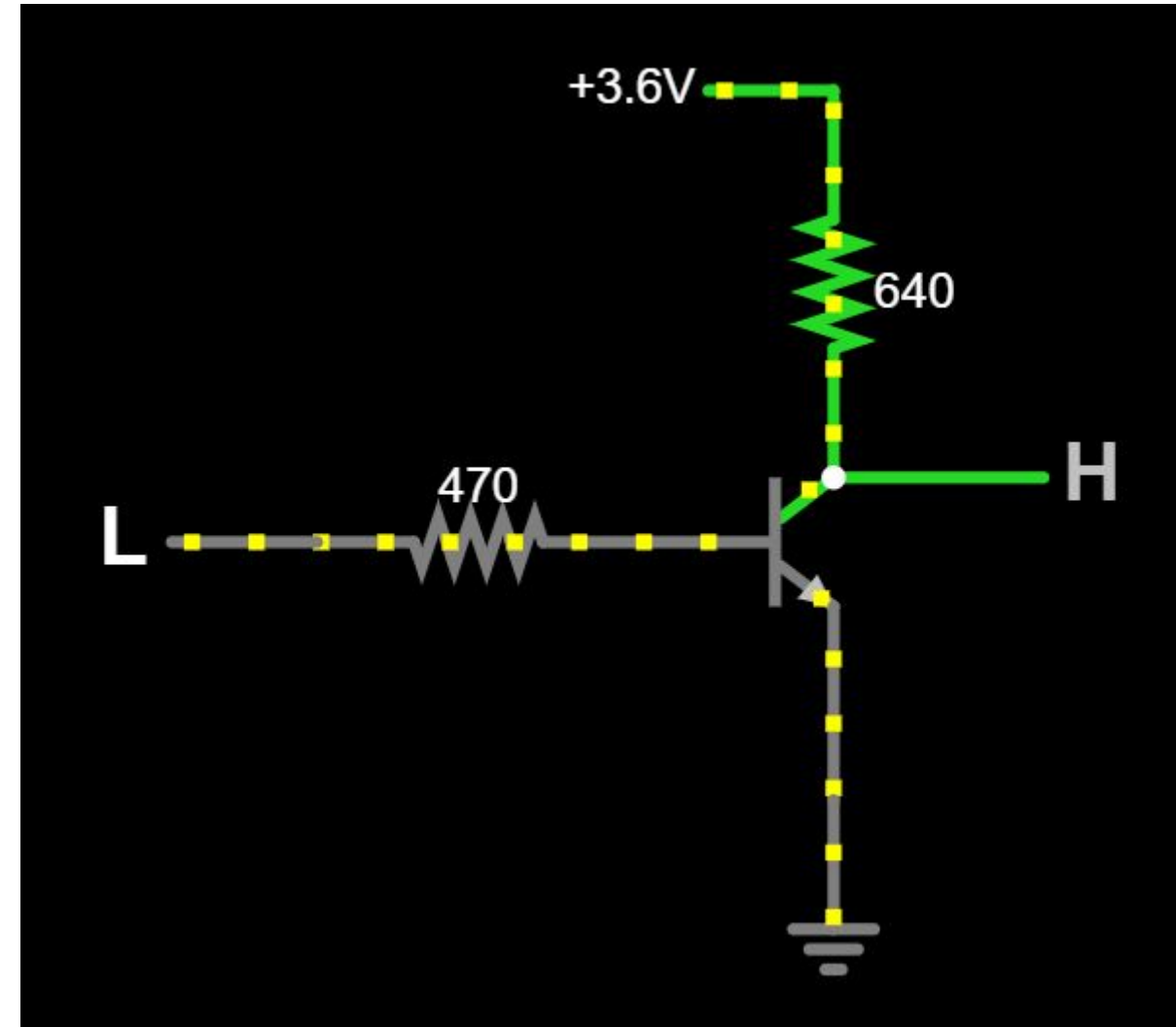
What is inside these transistors?

Diodes? FETs?

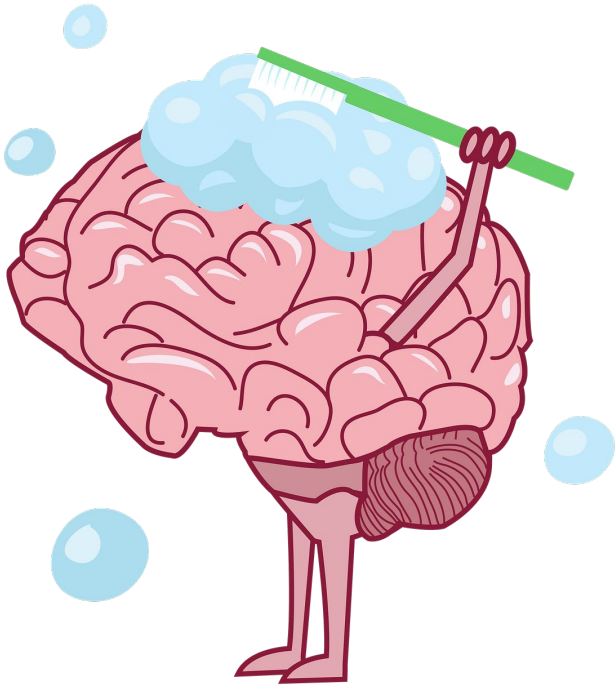
Let's get our basics brushed



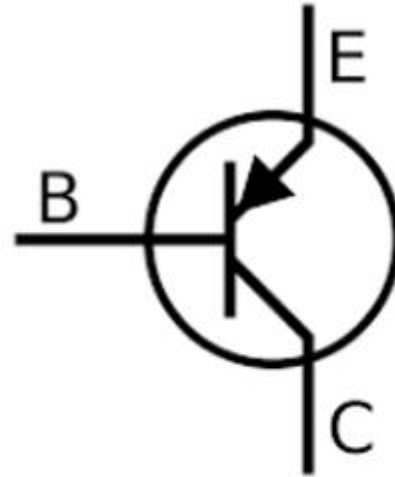
Transistors



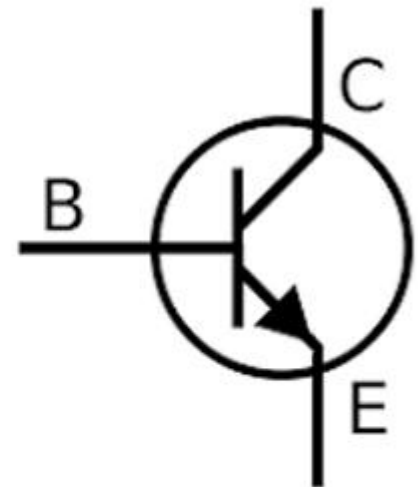
Let's get our basics brushed



Transistors



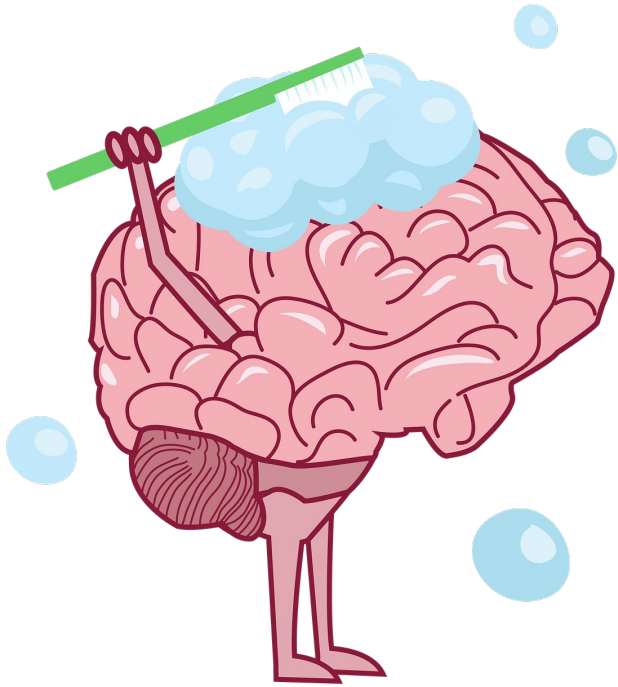
PNP BJT



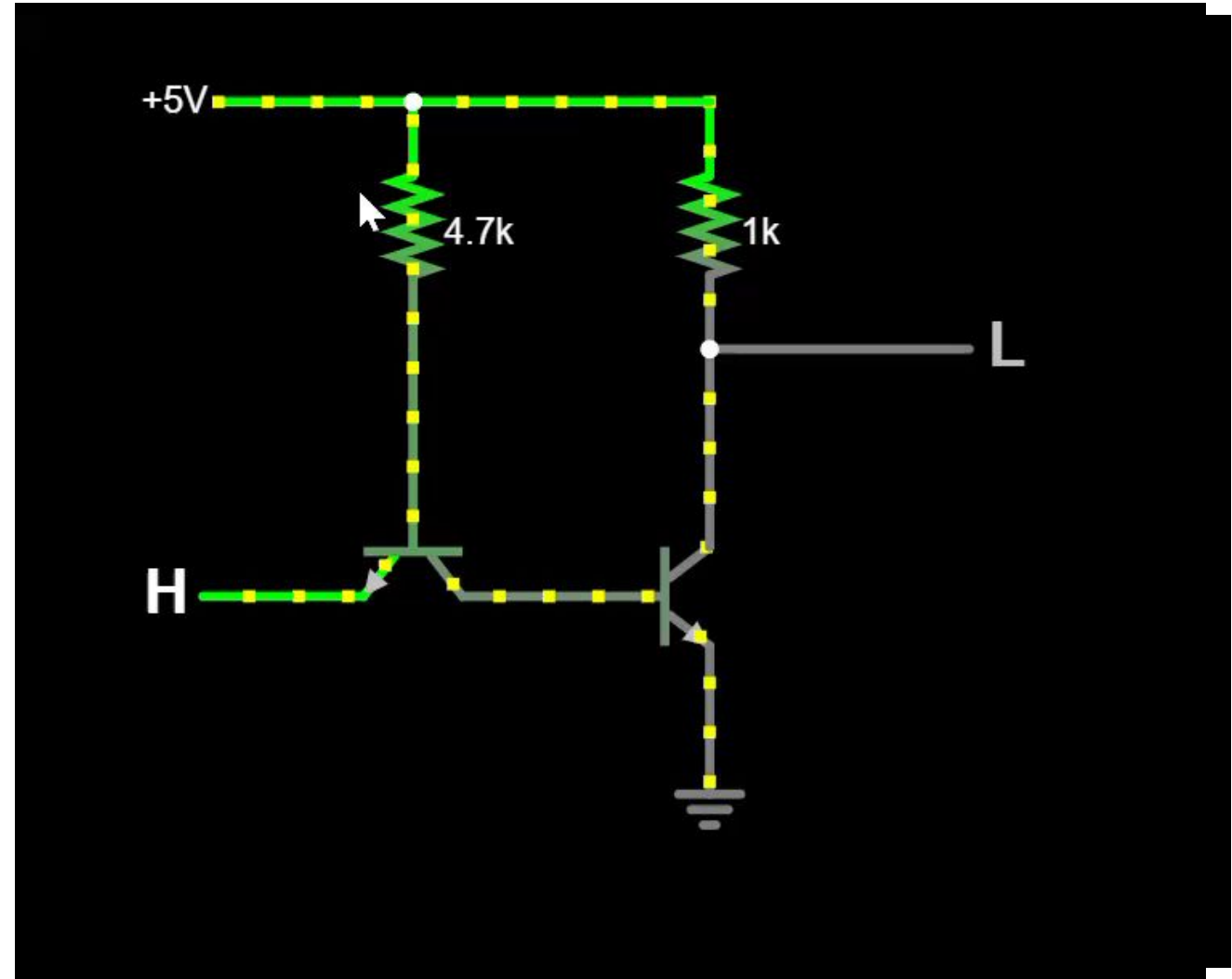
NPN BJT

Are transistors analog or Digital?

Let's get our basics brushed

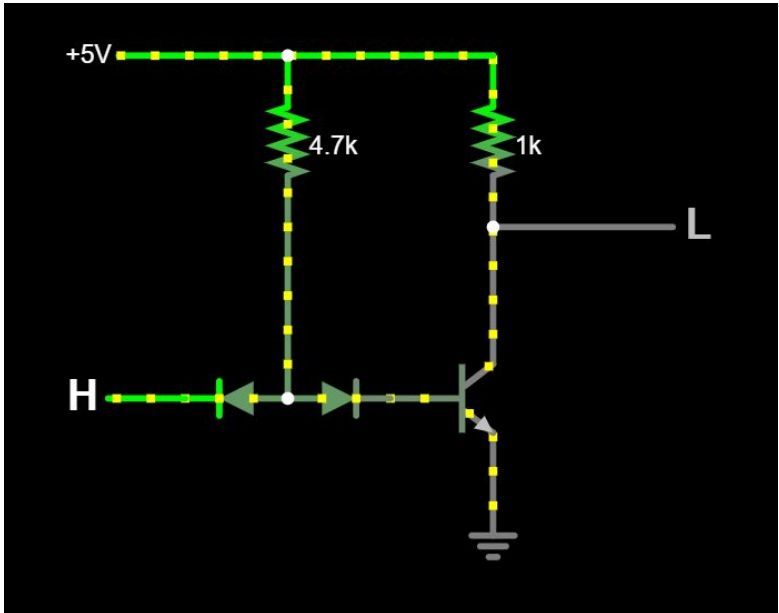
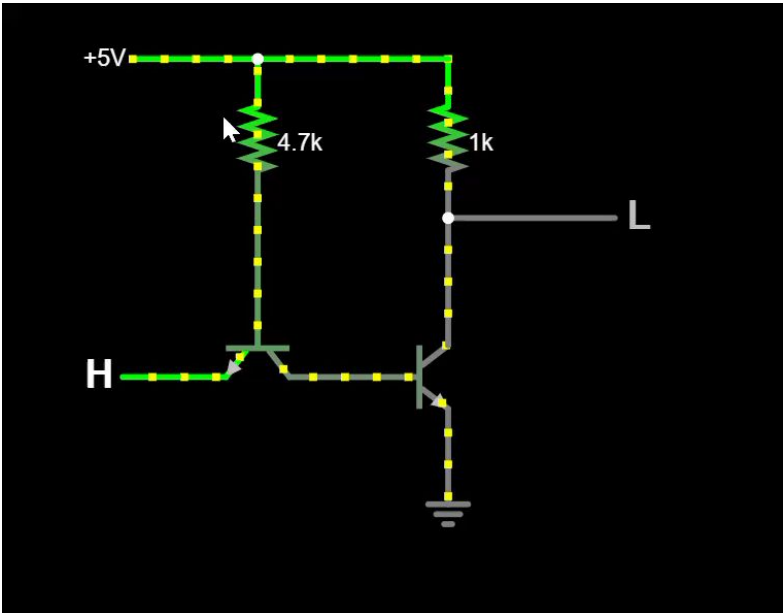
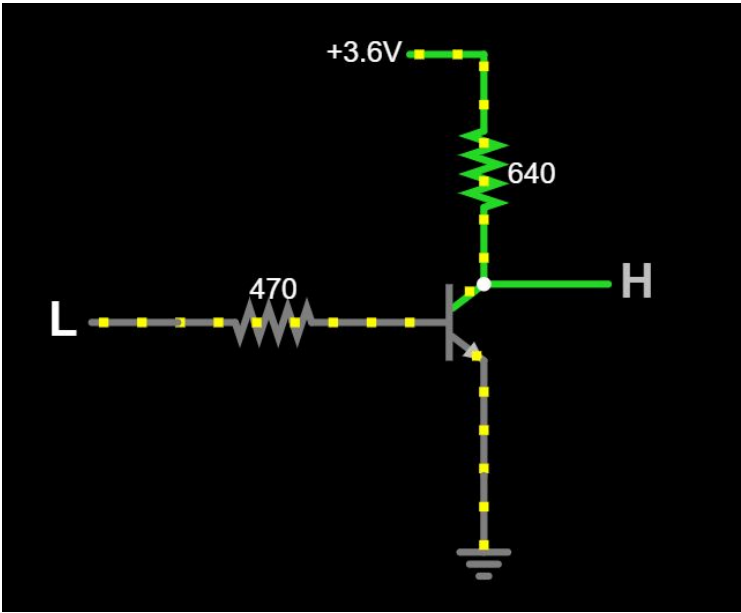


Transistors

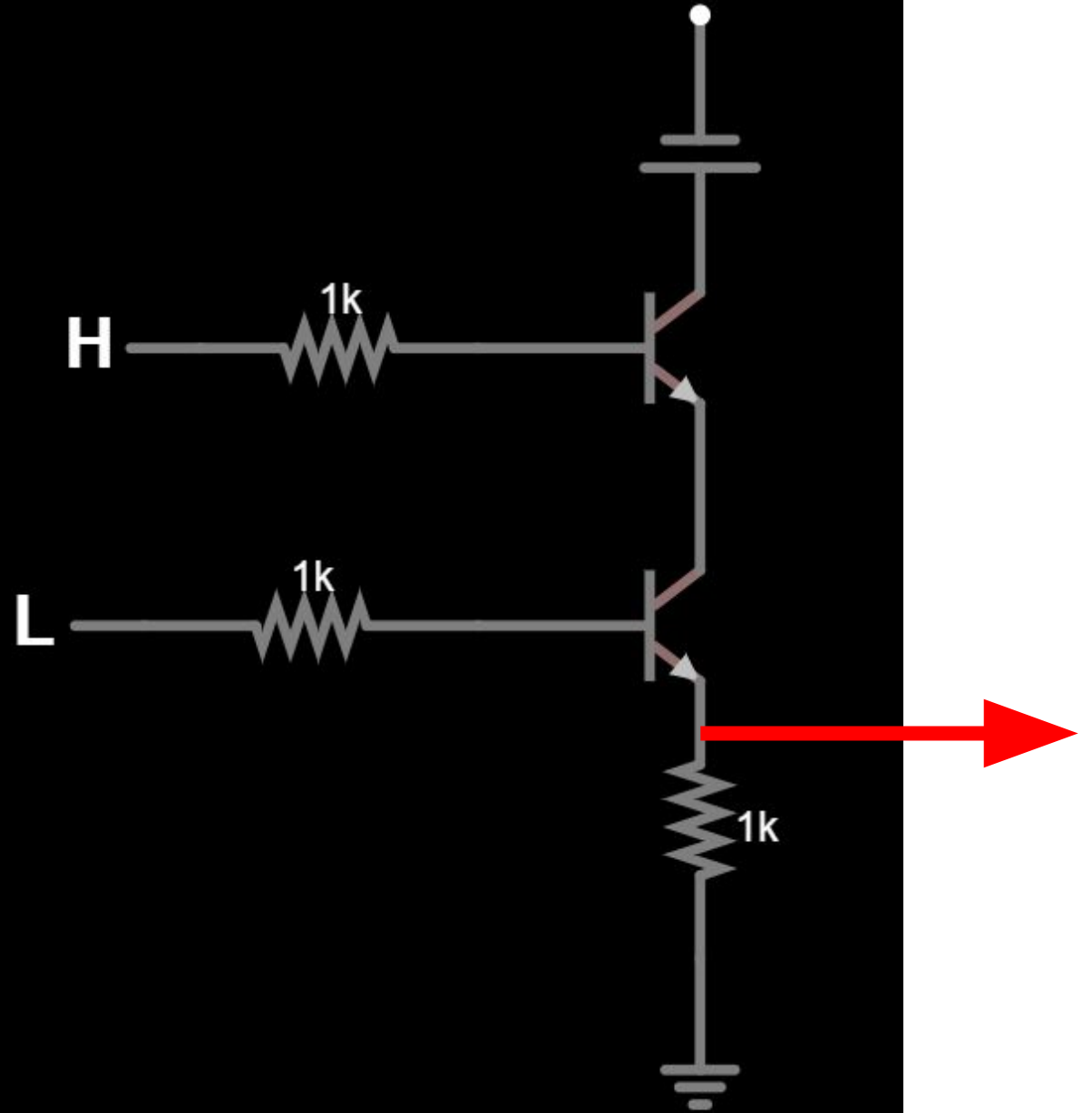


Differences in the above two is, one is RTL, another is TTL and then there is also another formation DTL

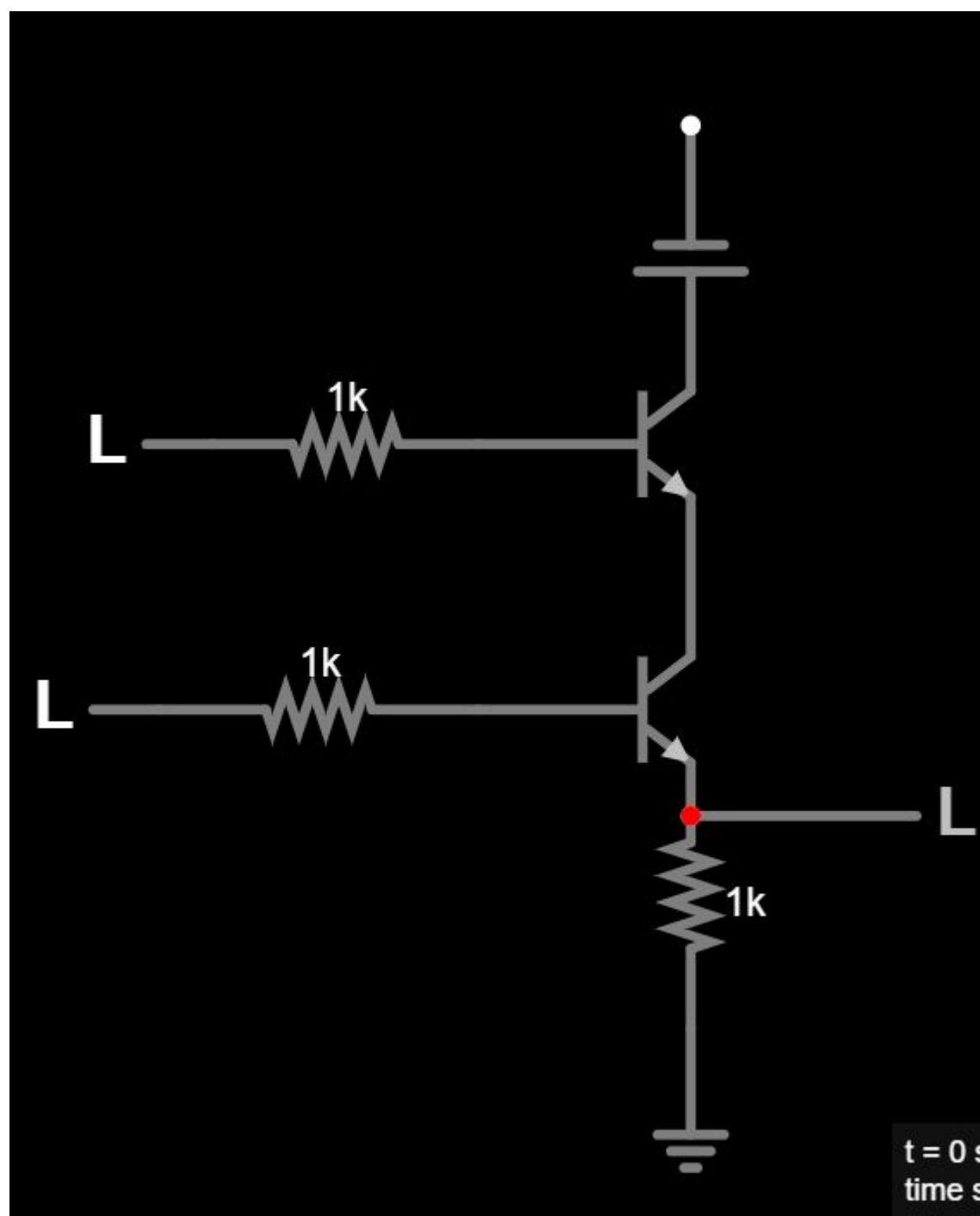
	RTL (Resistor-Transistor Logic)	TTL (Transistor-Transistor Logic)	DTL (Diode-Transistor Logic)
Components	Resistors and BJTs	BJTs for logic functions and amplification	Diodes for input logic, transistors for output amplification
Speed	Relatively slow	Faster than RTL and DTL	Slower than TTL but faster than RTL



**Analyze
And find the
GATE
configuration**



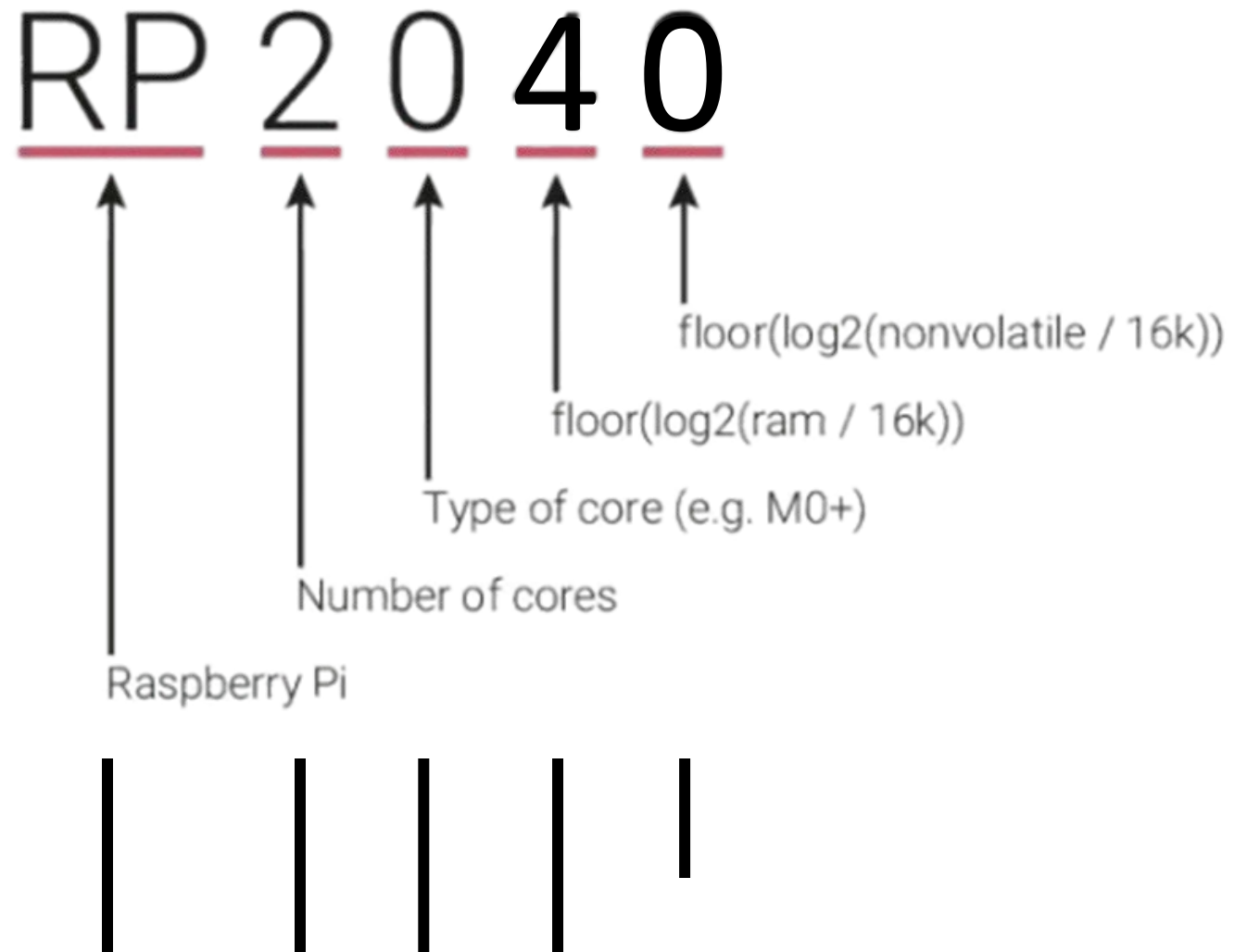
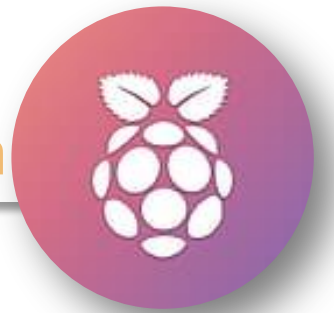
Analyze



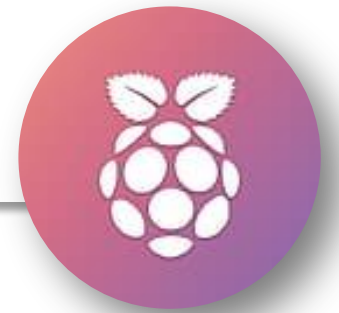


- **Introduction**
- **A system overview of the RP2040 chip**
- **Pin Descriptions**
- **GPIO Functions**

Introduction

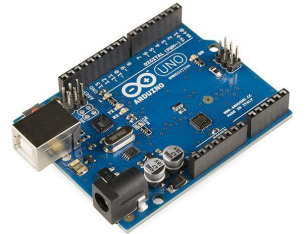


Summary

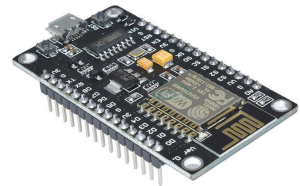


- Dual Cortex M0+ processor cores, up to **133MHz**
- 264kB of embedded SRAM in 6 banks
- 30 multifunction GPIO
- 6 dedicated IO for SPI Flash (supporting XIP)
- Dedicated hardware for commonly used peripherals
- Programmable IO for extended peripheral support
- 4 channel ADC with internal temperature sensor, 500ksps, 12-bit conversion
- USB 1.1 Host/Device

16MHz



160MHz



240MHz



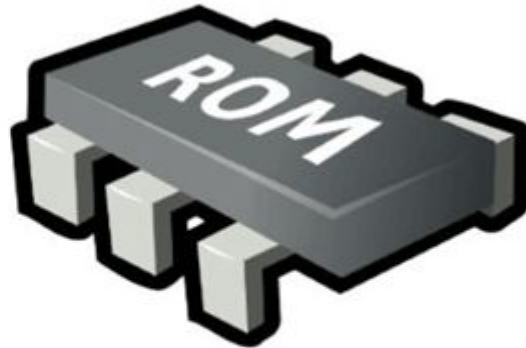
- 30 multifunction GPIO



- Dual Cortex M0+ processor cores, up to 133MHz

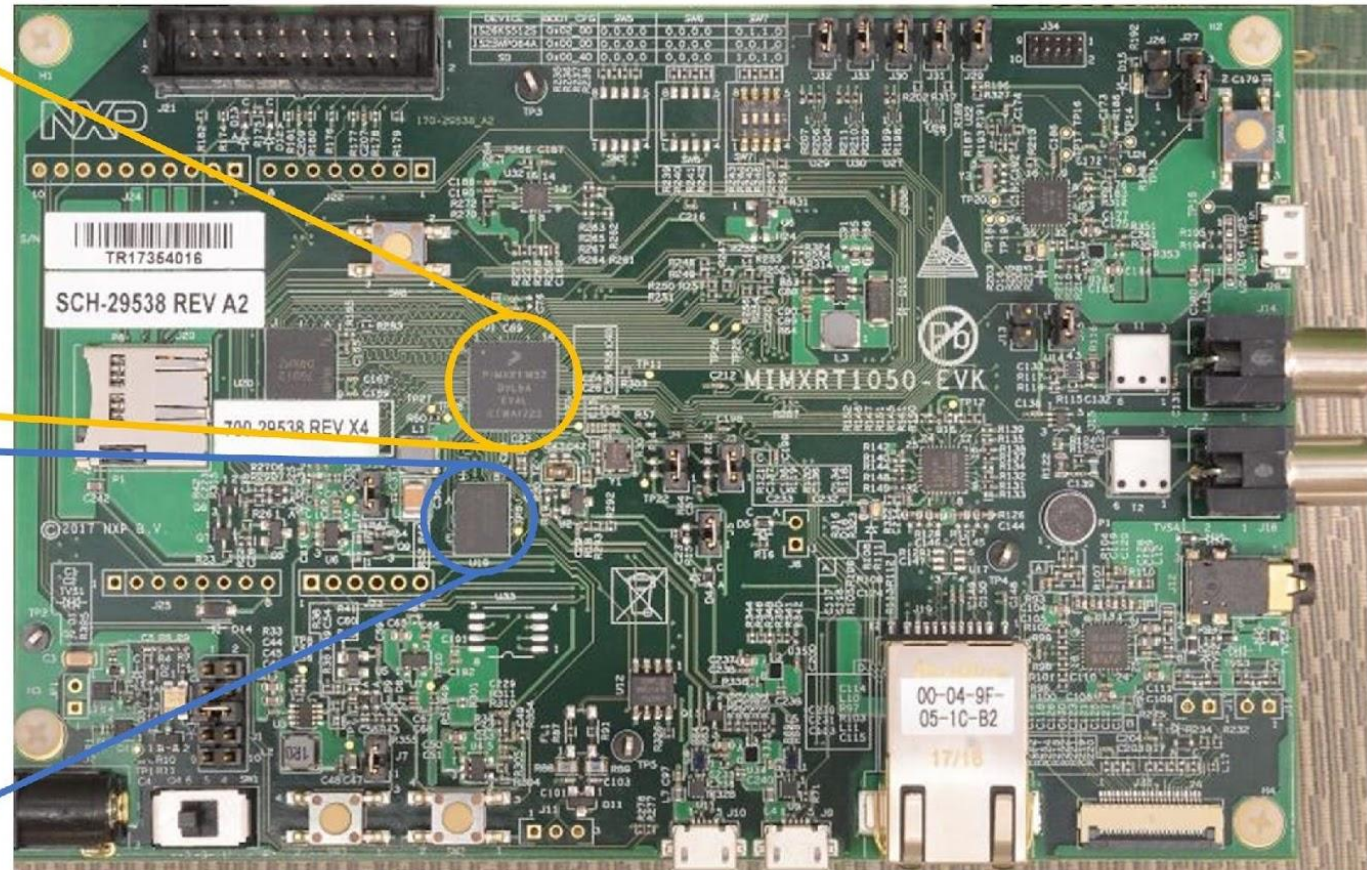
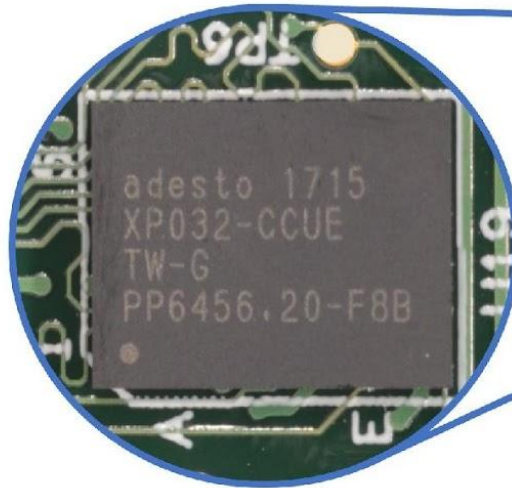


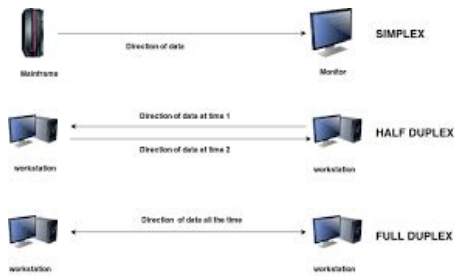
- 264kB of embedded SRAM in 6 banks



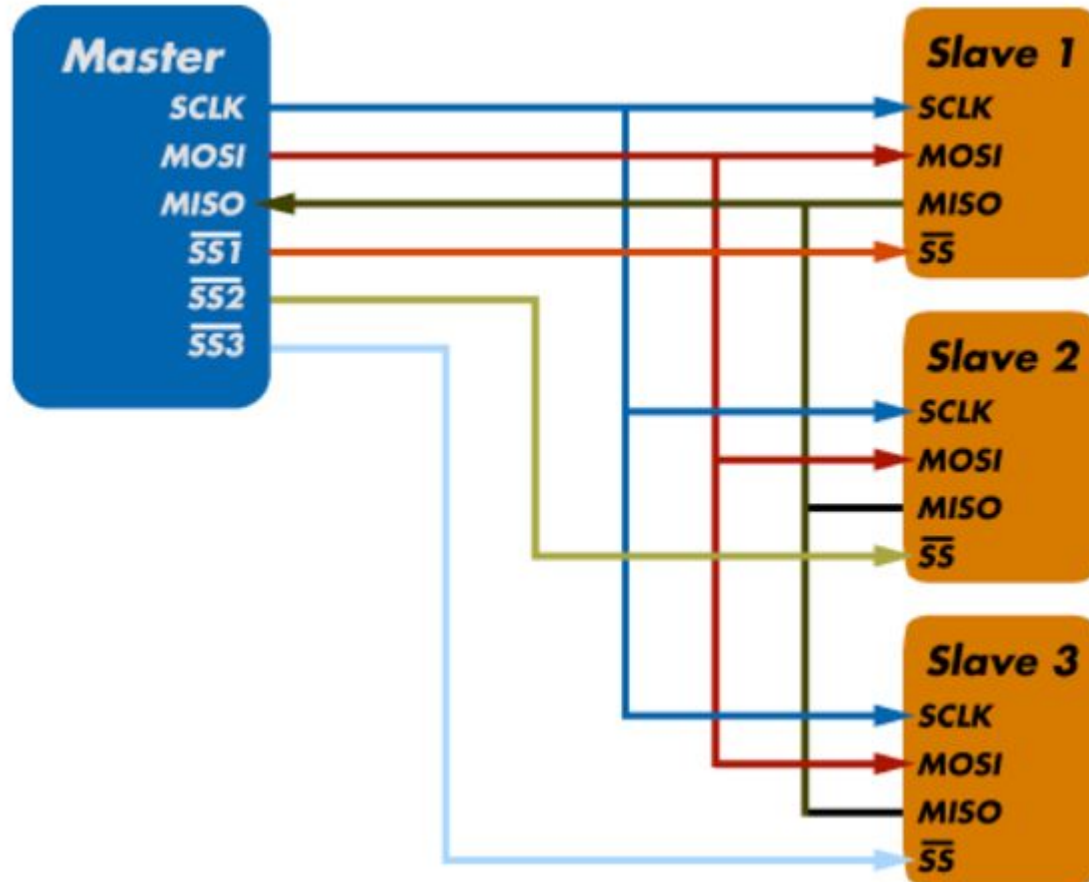
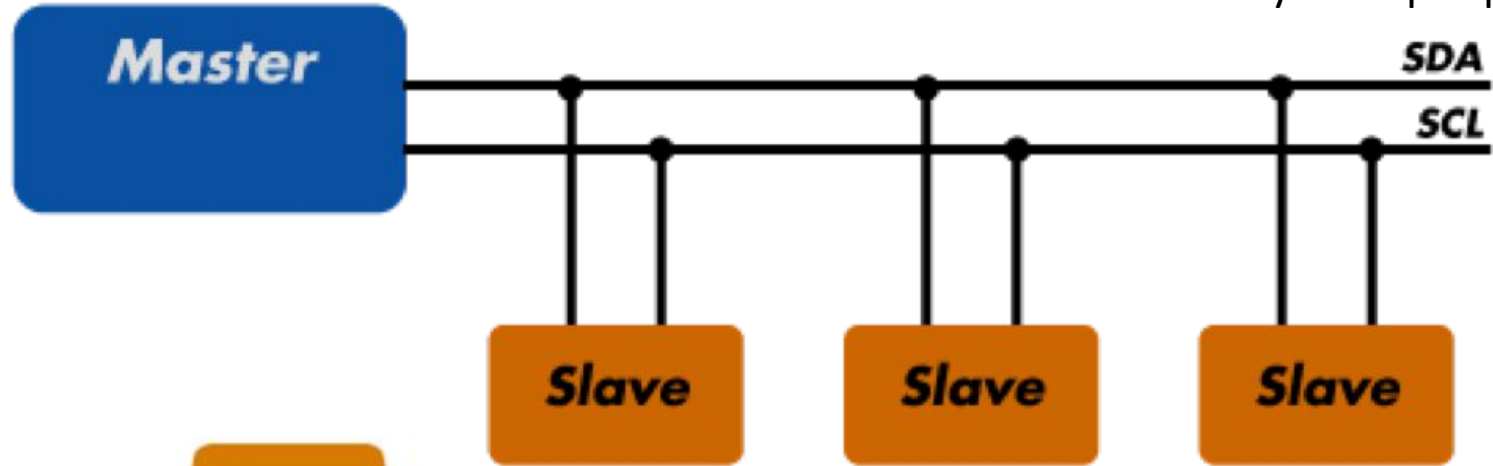
RAM Vs ROM

- 6 dedicated IO for SPI Flash (supporting XIP)

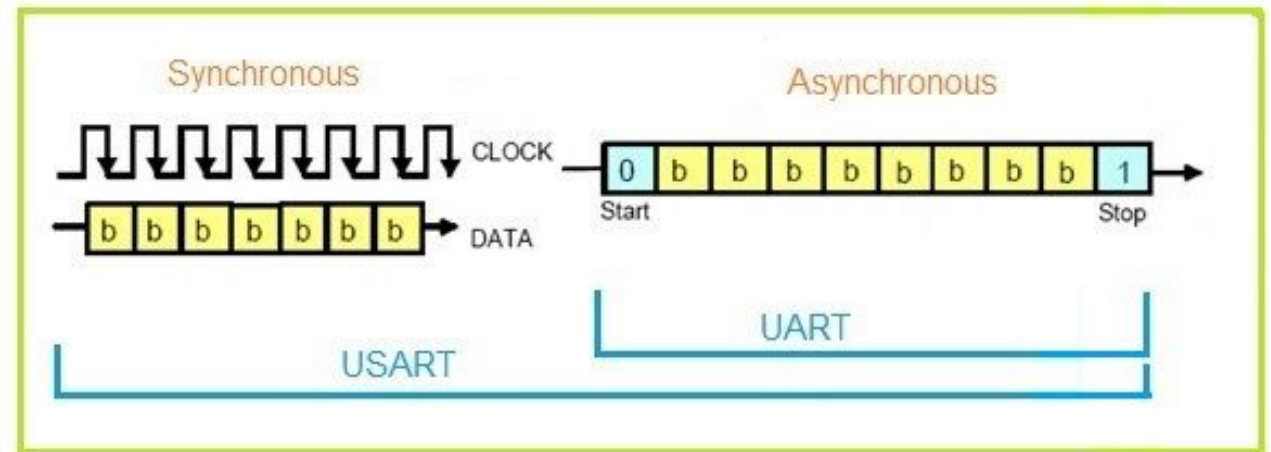




- Dedicated hardware for commonly used peripherals



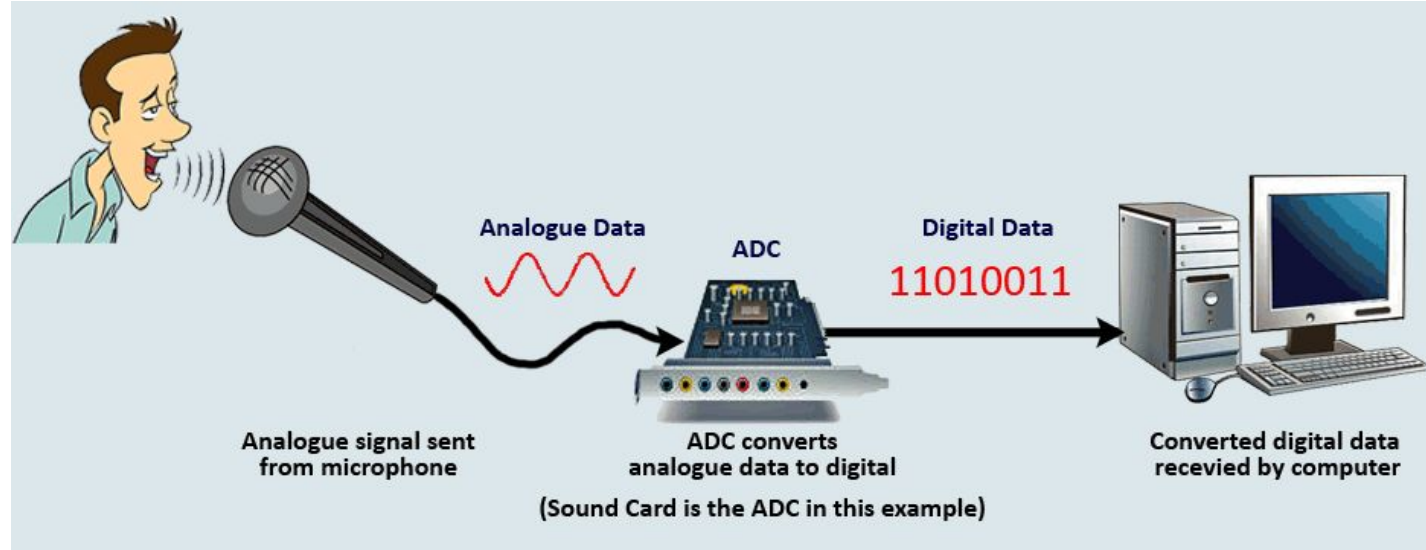
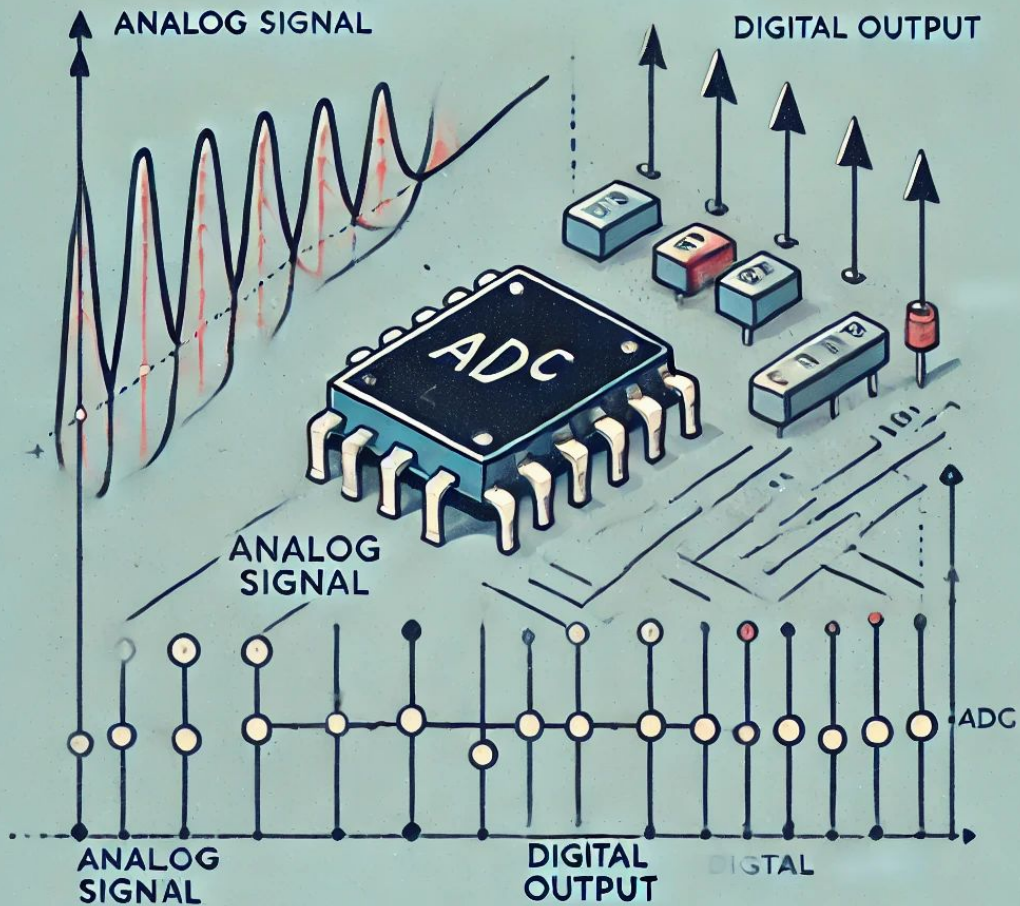
I2C vs SPI vs UART



- Programmable IO for extended peripheral support

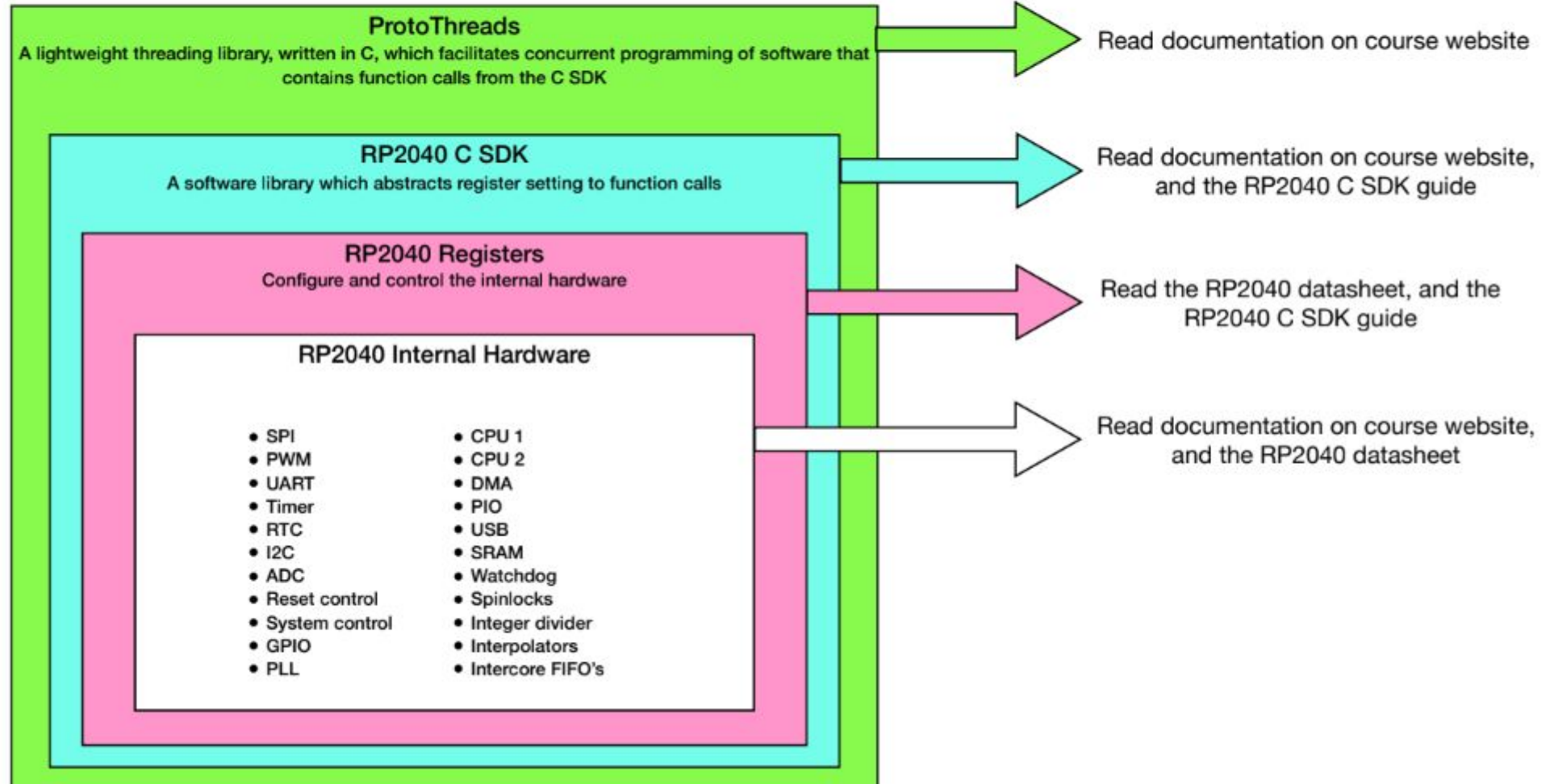


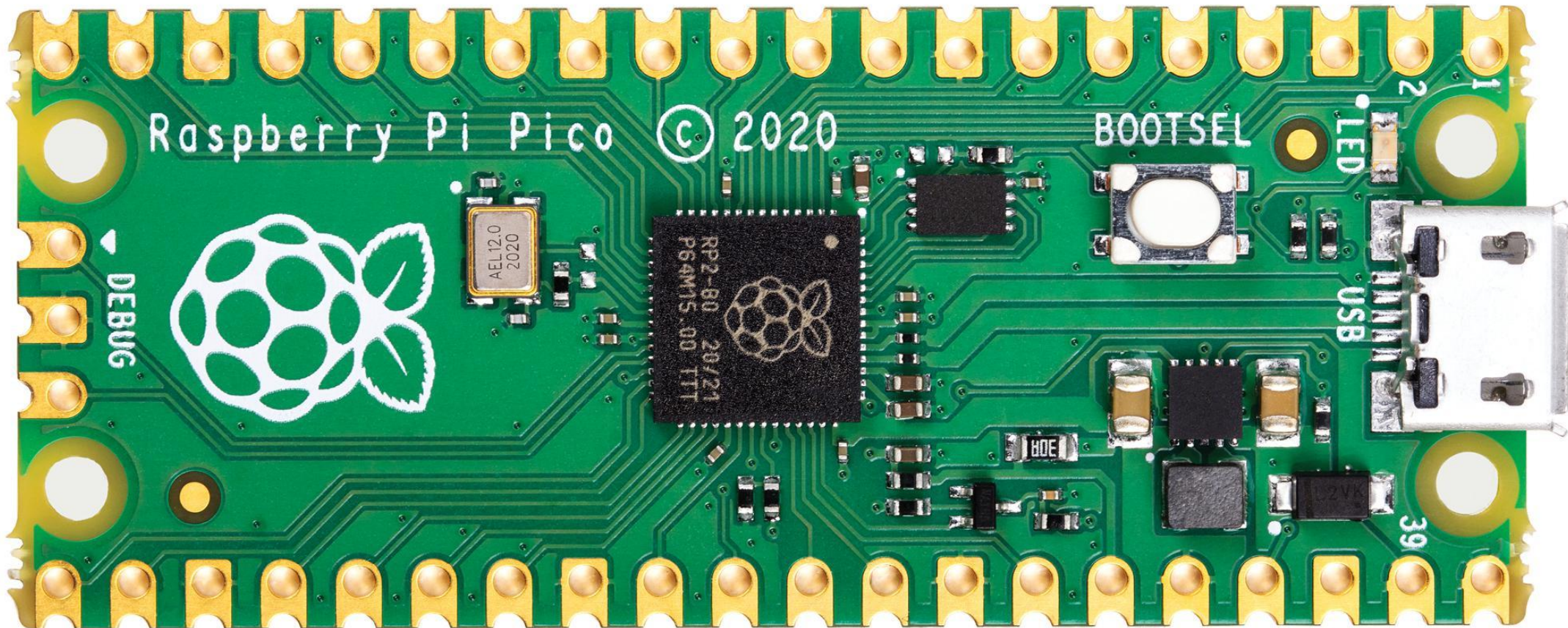
ANALOG TO DIGITAL



- 4 channel ADC with internal temperature sensor, 500ksps, 12-bit conversion

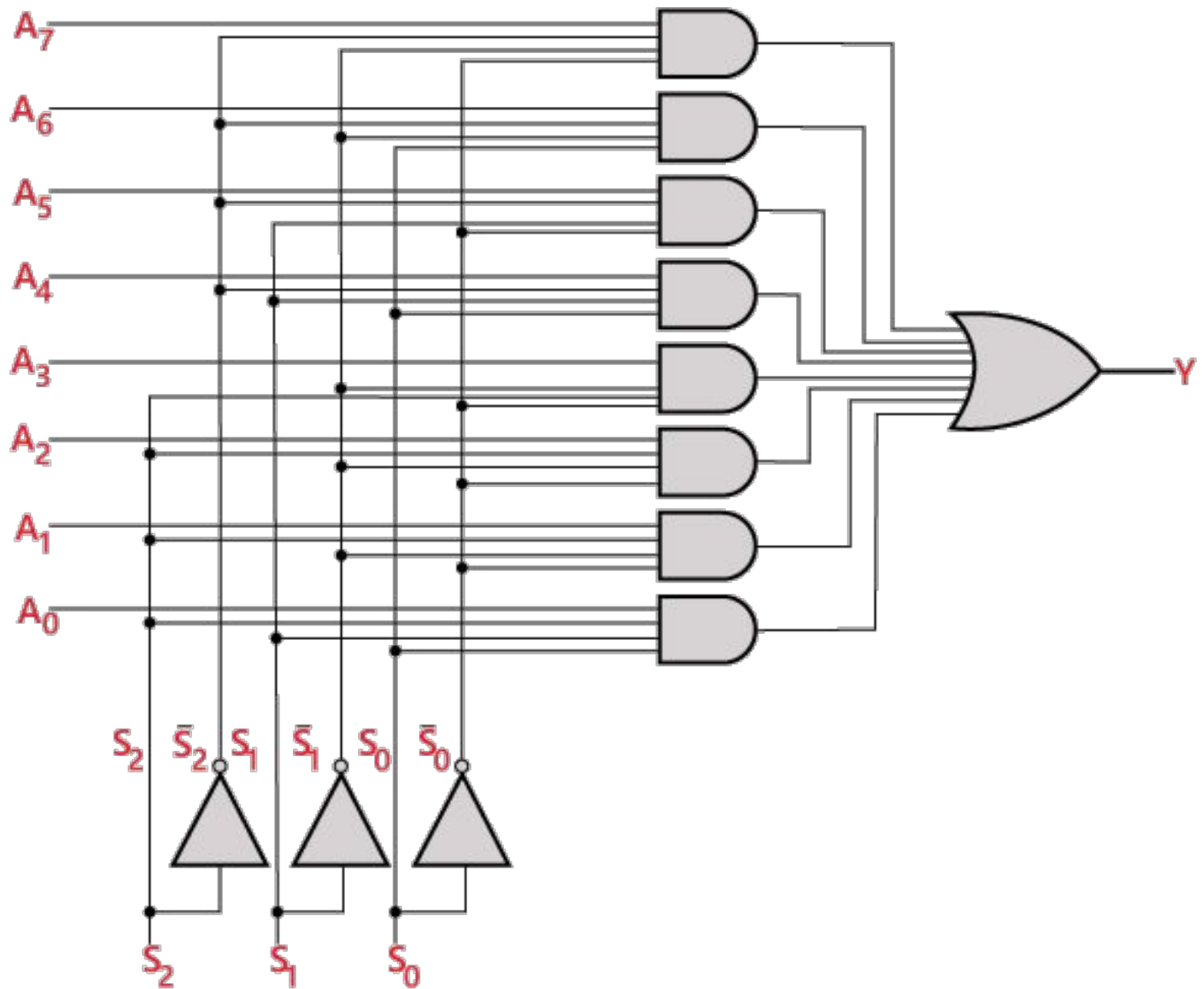
Software/hardware levels of abstraction





Was ist das?

Und warum brauchen wir das?



How these registers are used in program?

	1	2	3	4	5
A					
B					
C		XXXX			
D					
E					

How these registers are used in program?

2.3.1.7. List of Registers

The SIO registers start at a base address of `0xd0000000` (defined as `SIO_BASE` in SDK).

Offset	Name	Info
0x000	<code>CPUID</code>	Processor core identifier
0x004	<code>GPIO_IN</code>	Input value for <code>GPIO</code> pins
0x008	<code>GPIO_HI_IN</code>	Input value for QSPI pins
0x010	<code>GPIO_OUT</code>	<code>GPIO</code> output value
0x014	<code>GPIO_OUT_SET</code>	<code>GPIO</code> output value set
0x018	<code>GPIO_OUT_CLR</code>	<code>GPIO</code> output value clear
0x01c	<code>GPIO_OUT_XOR</code>	<code>GPIO</code> output value XOR
0x020	<code>GPIO_OE</code>	<code>GPIO</code> output enable
0x024	<code>GPIO_OE_SET</code>	<code>GPIO</code> output enable set
0x028	<code>GPIO_OE_CLR</code>	<code>GPIO</code> output enable clear

How these registers are used in program?

SIO: **GPIO_IN** Register

Offset: 0x004

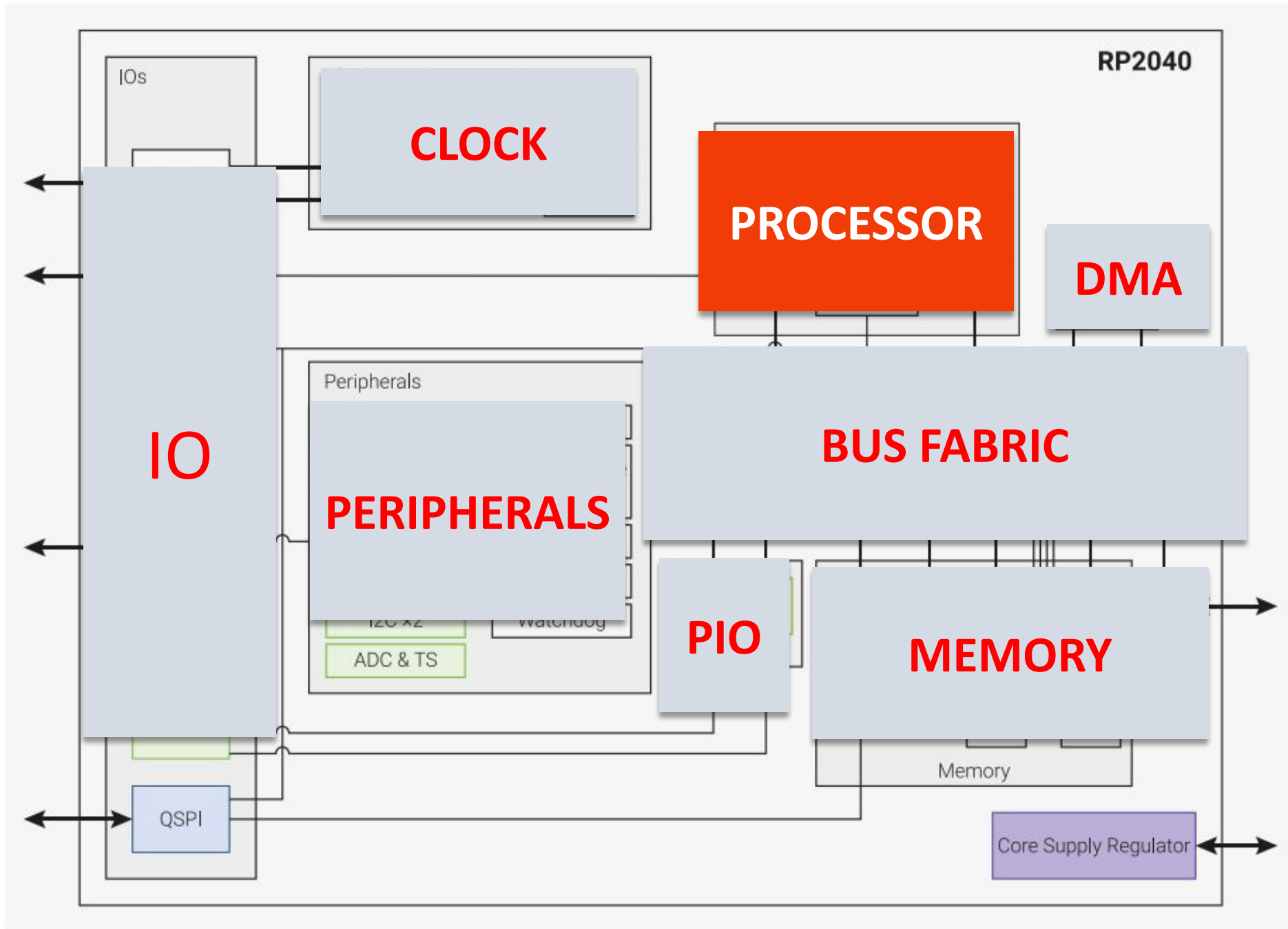
Description

Input value for **GPIO** pins

Bits	Description	Type	Reset
31:30	Reserved.	-	-
29:0	Input value for GPIO 0...29	RO	0x00000000

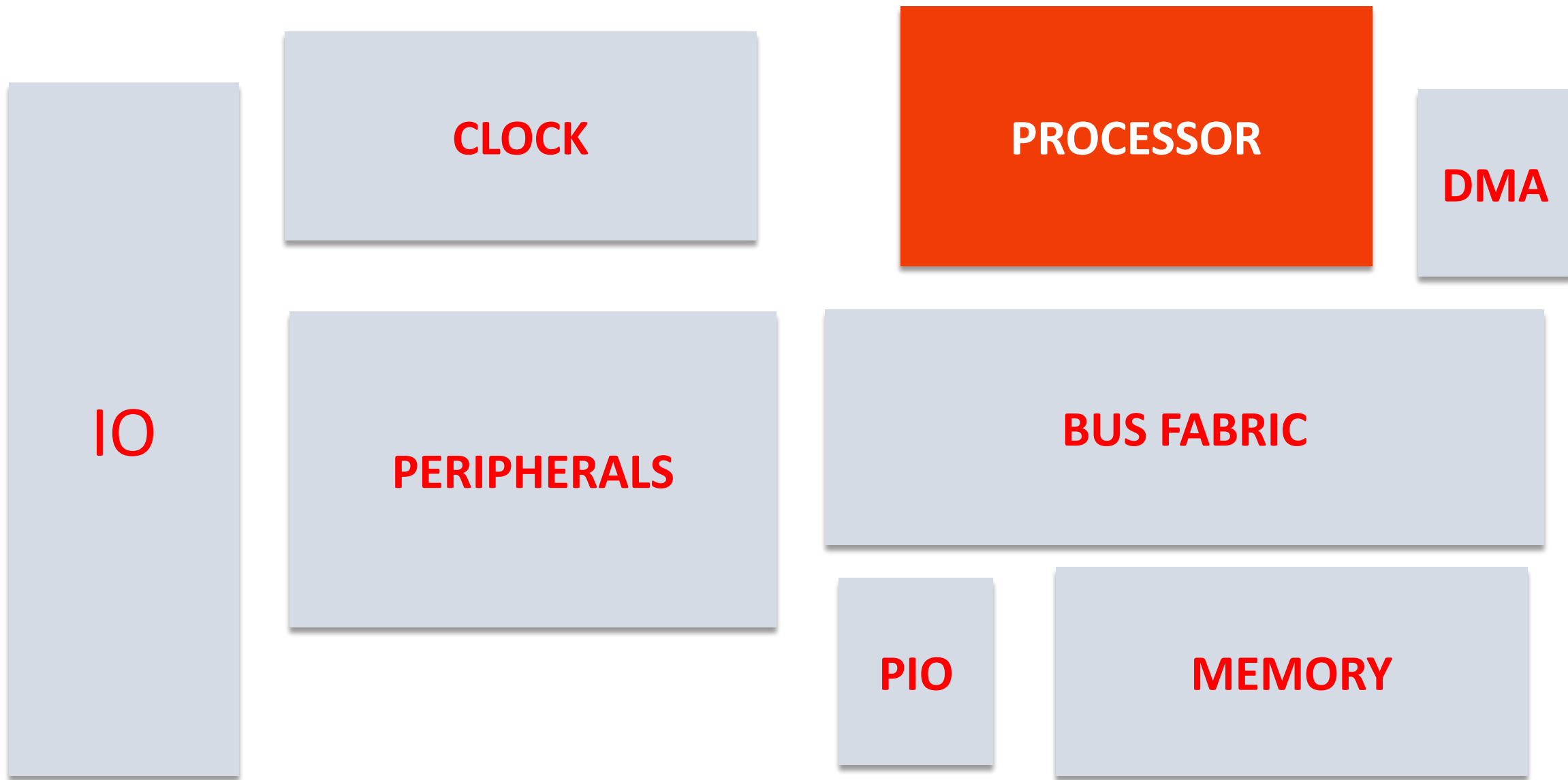
	04																															
0xd000000	3 1	3 0	2 9	2 8	2 7	2 6	2 5	2 4	2 3	2 2	2 1	2 0	1 9	1 8	1 7	1 6	1 5	1 4	1 3	1 2	1 1	1 0	9	8	7	6	5	4	3	2	1	0
																															1	0

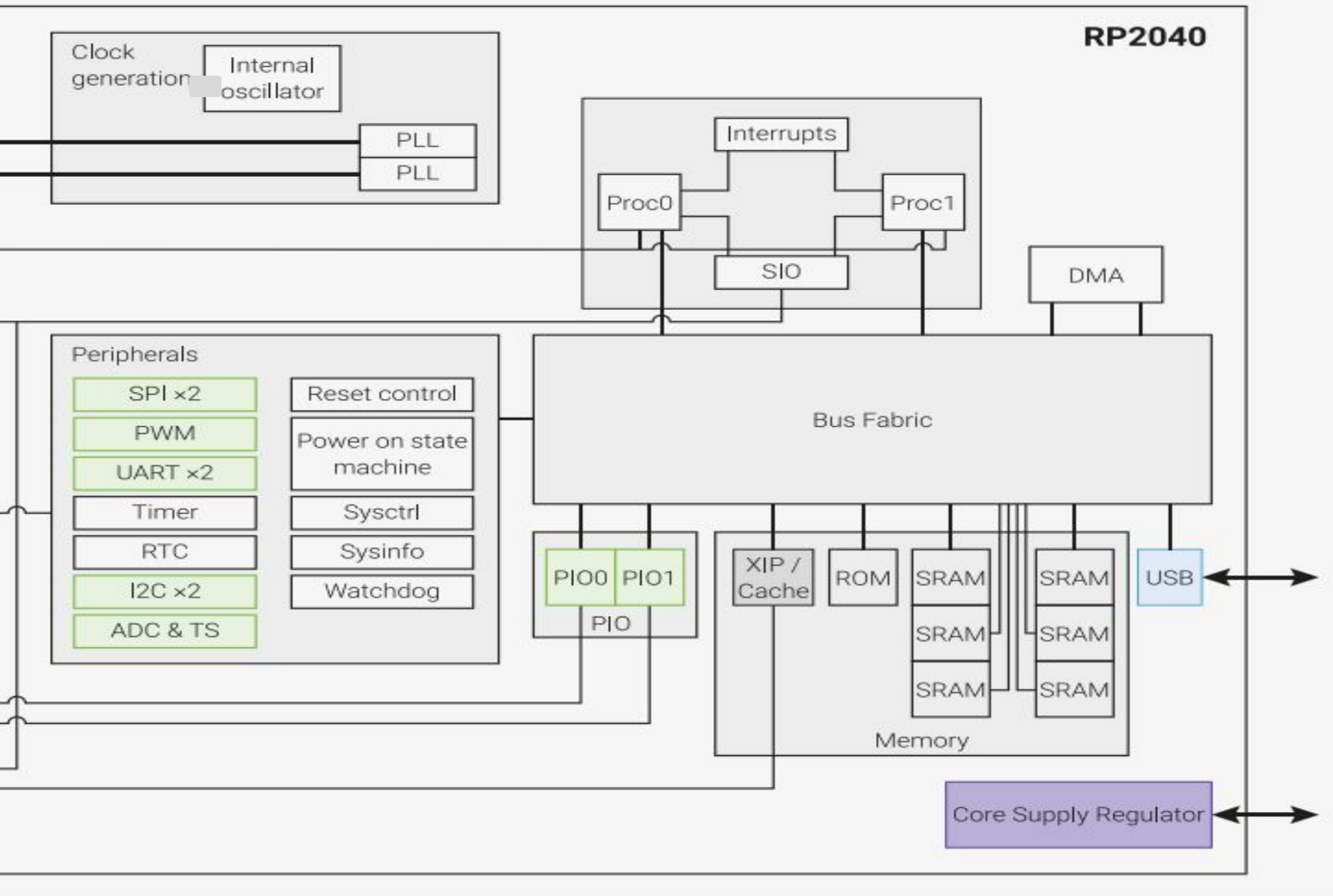
A system overview
of the RP2040 chip



RP2040







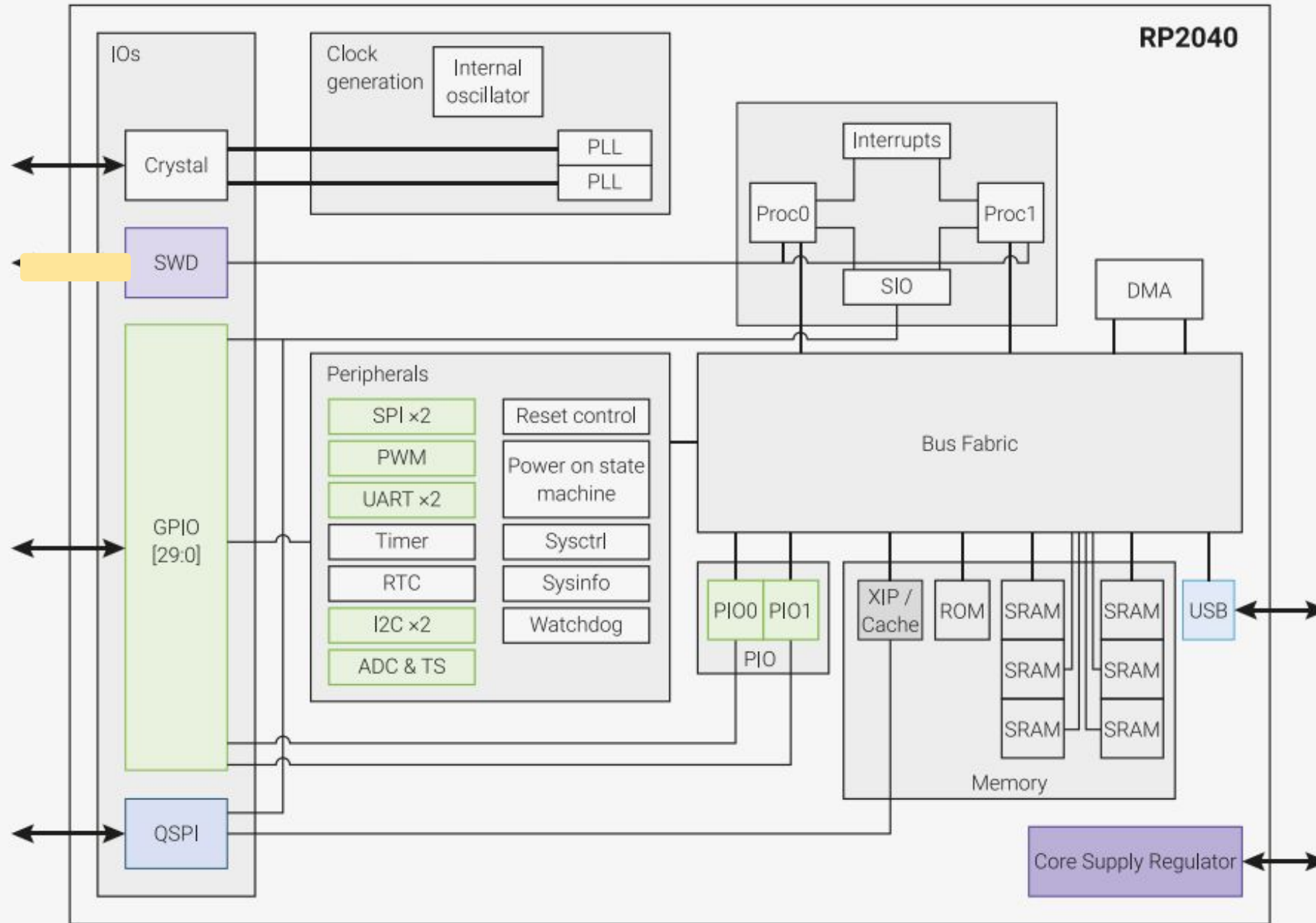
RP2040

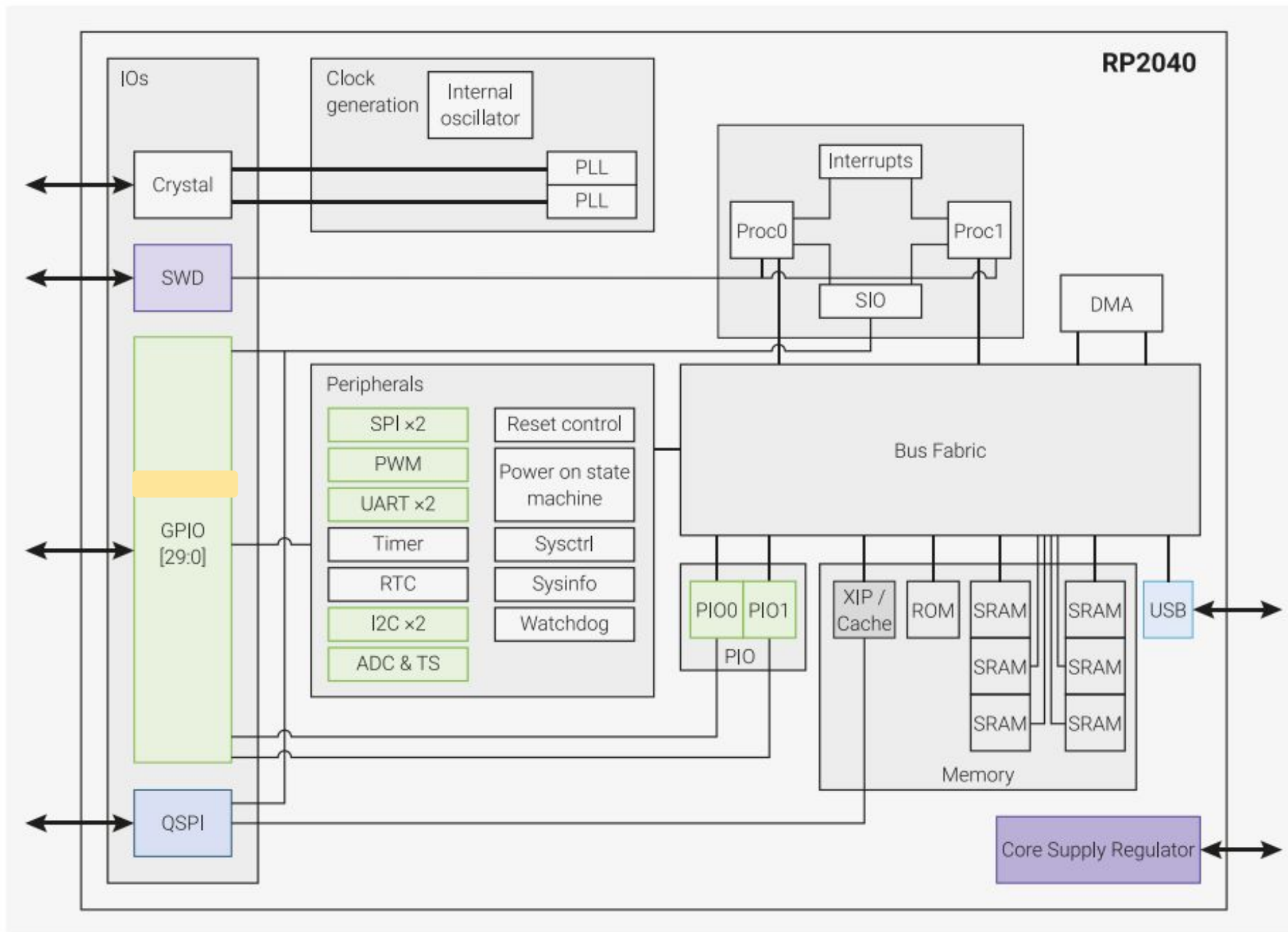
Crystal is used to generate the clock pulse for the system.

In addition there is clock generation to enable the different clock values. Further PLL is provided to support the operation

RP2040

Access to the internal Serial Wire Debug multi-drop bus. Provides debug access to both processors, and can be used to download code

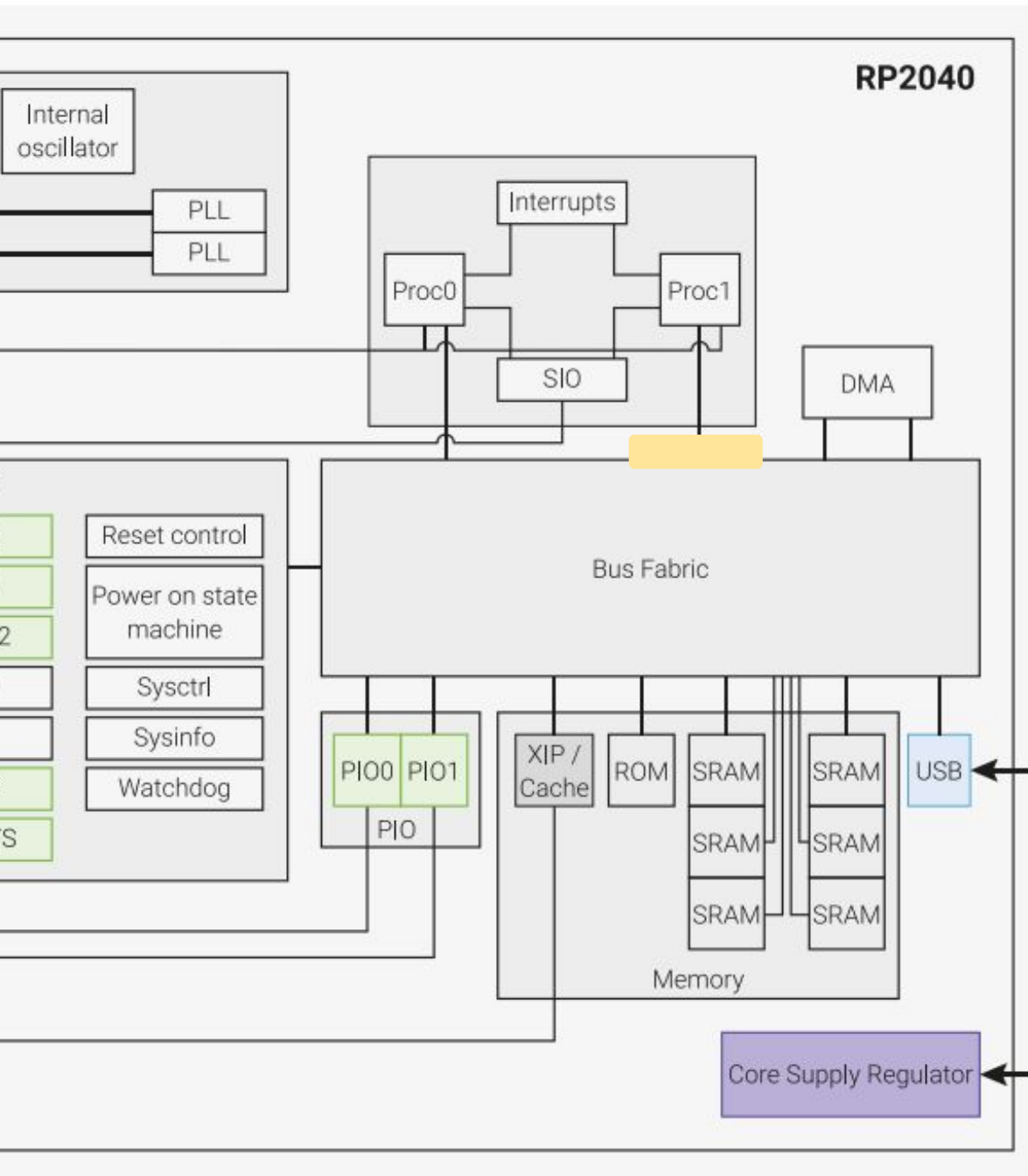




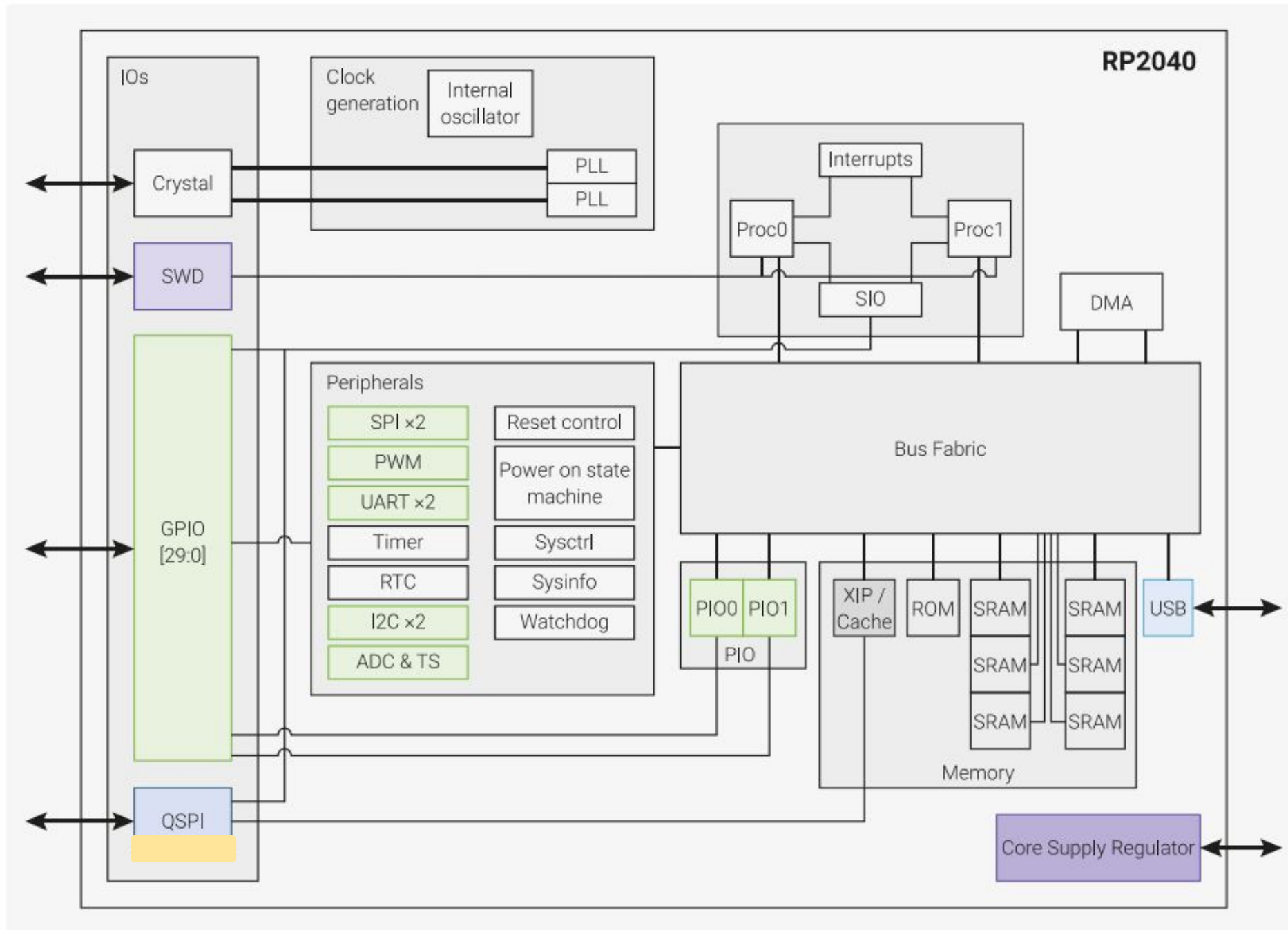
RP2040

General-purpose digital input and output. RP2040 can connect one of a number of internal peripherals to each GPIO, or control GPIOs directly from software.

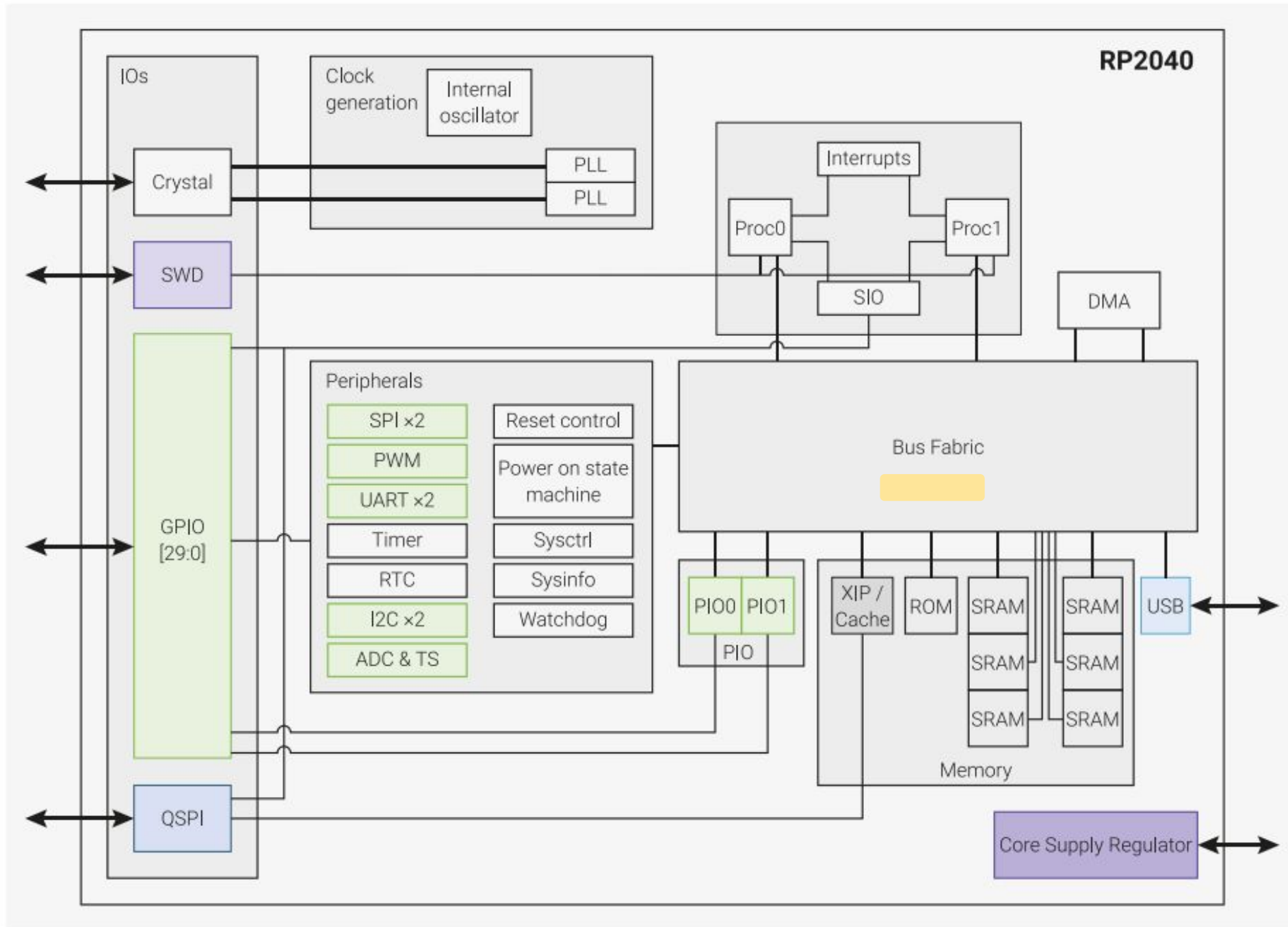
26-29 - ADC



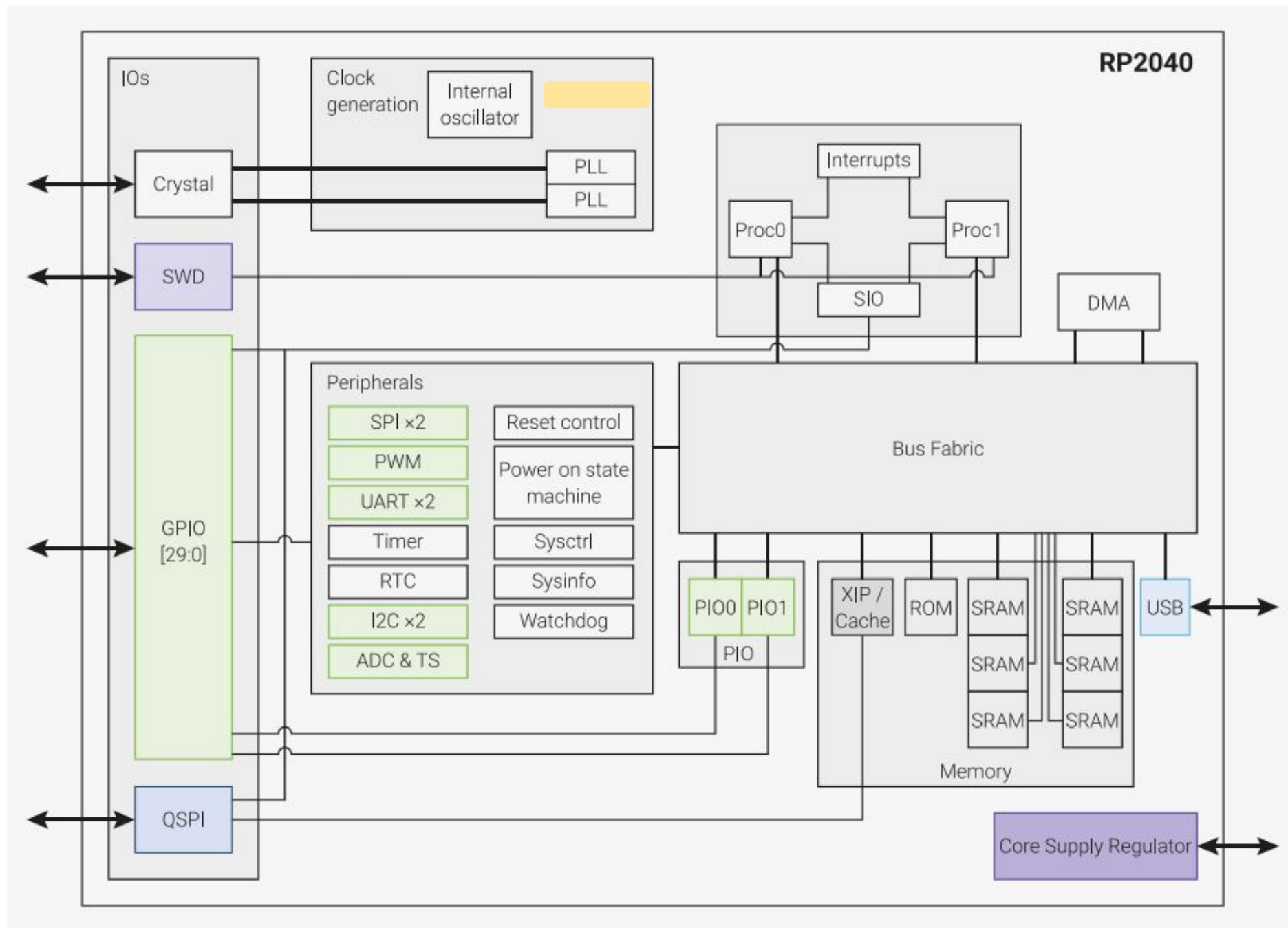
- **ControlReset** : Allows resetting the whole chip or a specific block to recover from faults.
- **Power-On State Machine**: Initializes clocks and configures the boot sequence during power-up.
- **SYSCTRL**: Manages system-level settings like clock configuration and subsystem enable/disable.
- **SYSINFO**: Provides read-only chip identification and version information registers.
- **Watchdog**: Resets the chip or triggers events if the system is not responding.



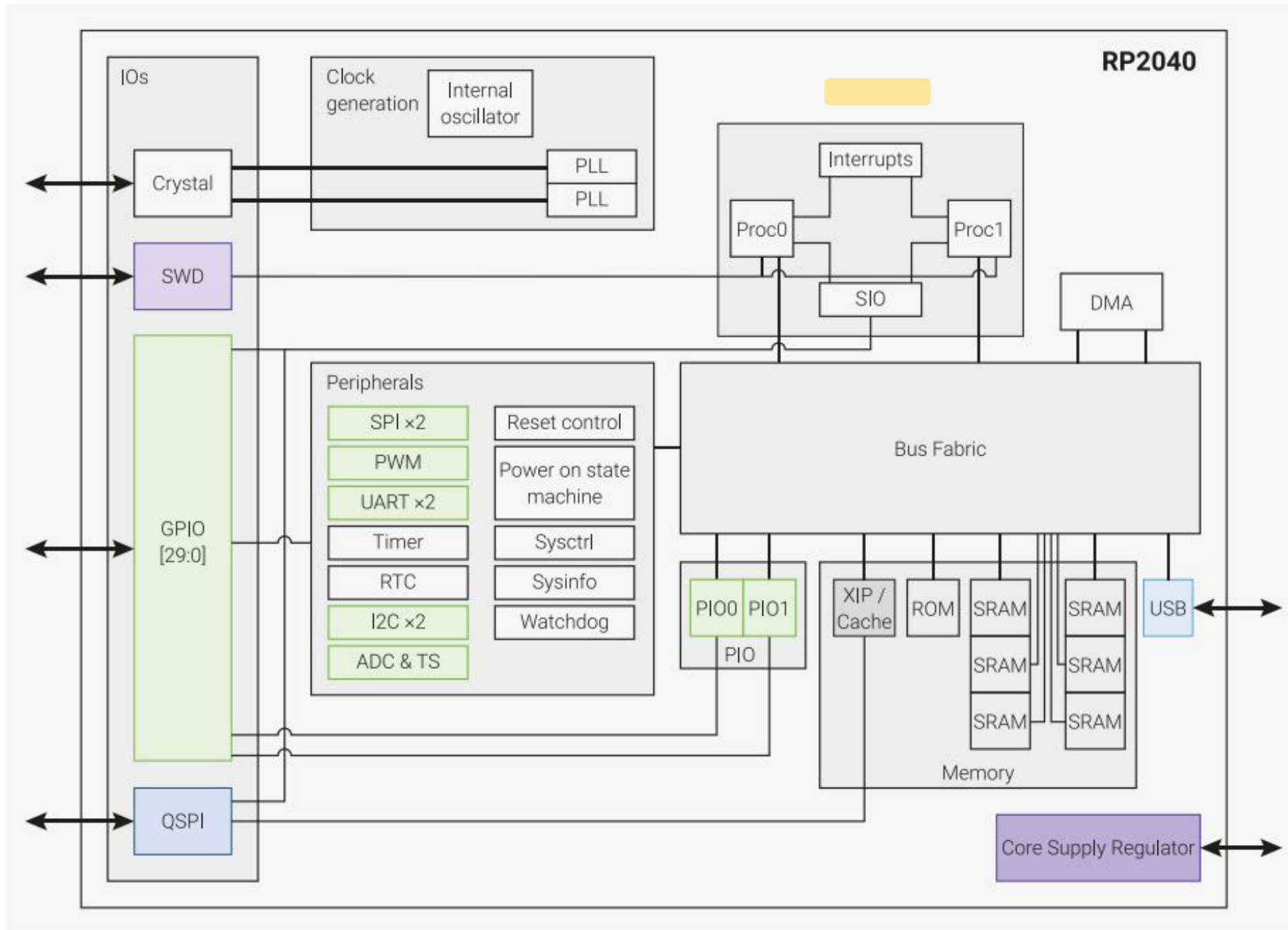
QSPI is a high-speed serial interface that uses four data lines to transfer data between a microcontroller and external flash memory. It enables faster read/write operations than standard SPI, often supporting execute-in-place (XIP) so code can run directly from external flash.



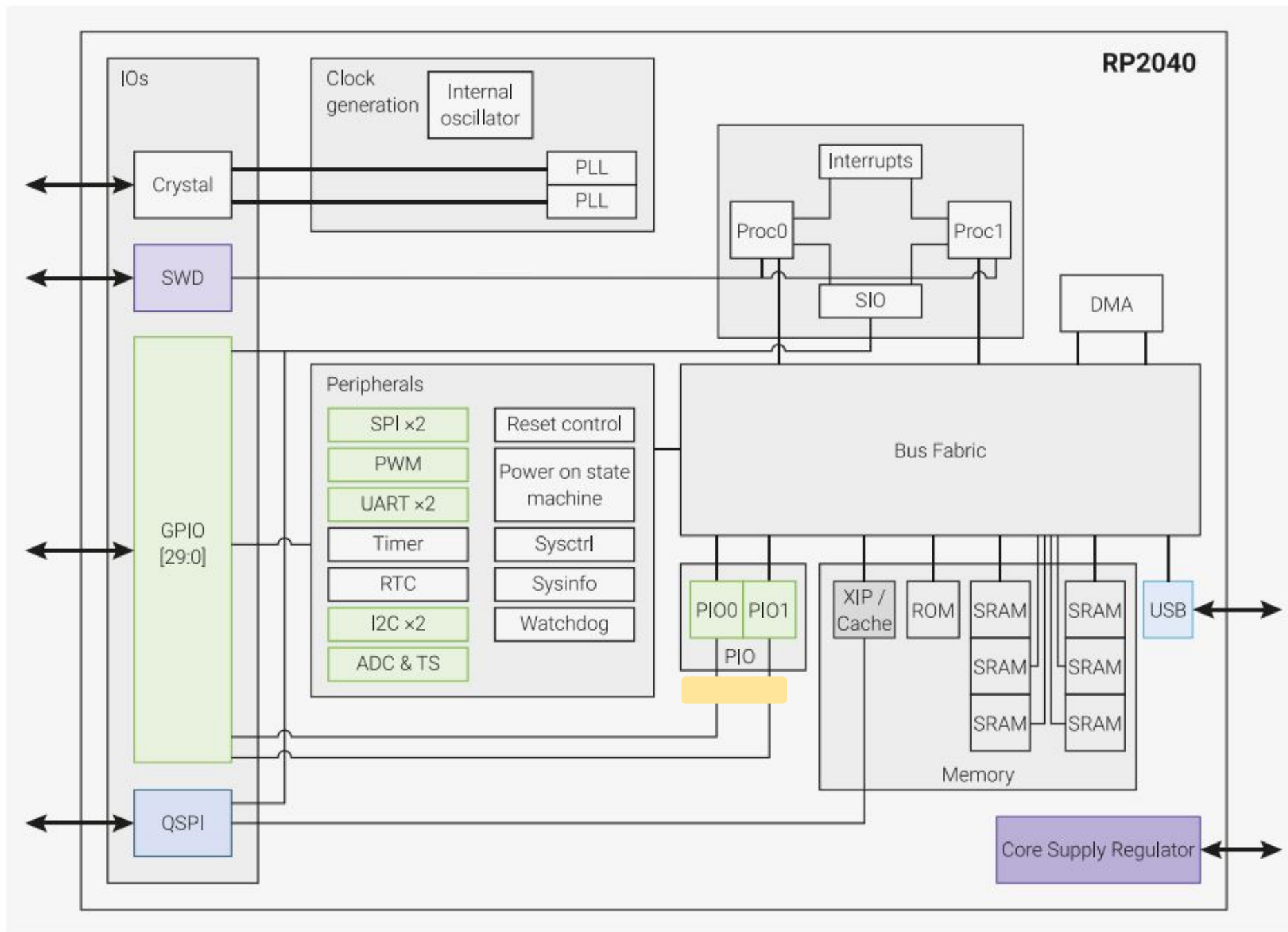
This is the structure which controls the bus over the regions of the system.



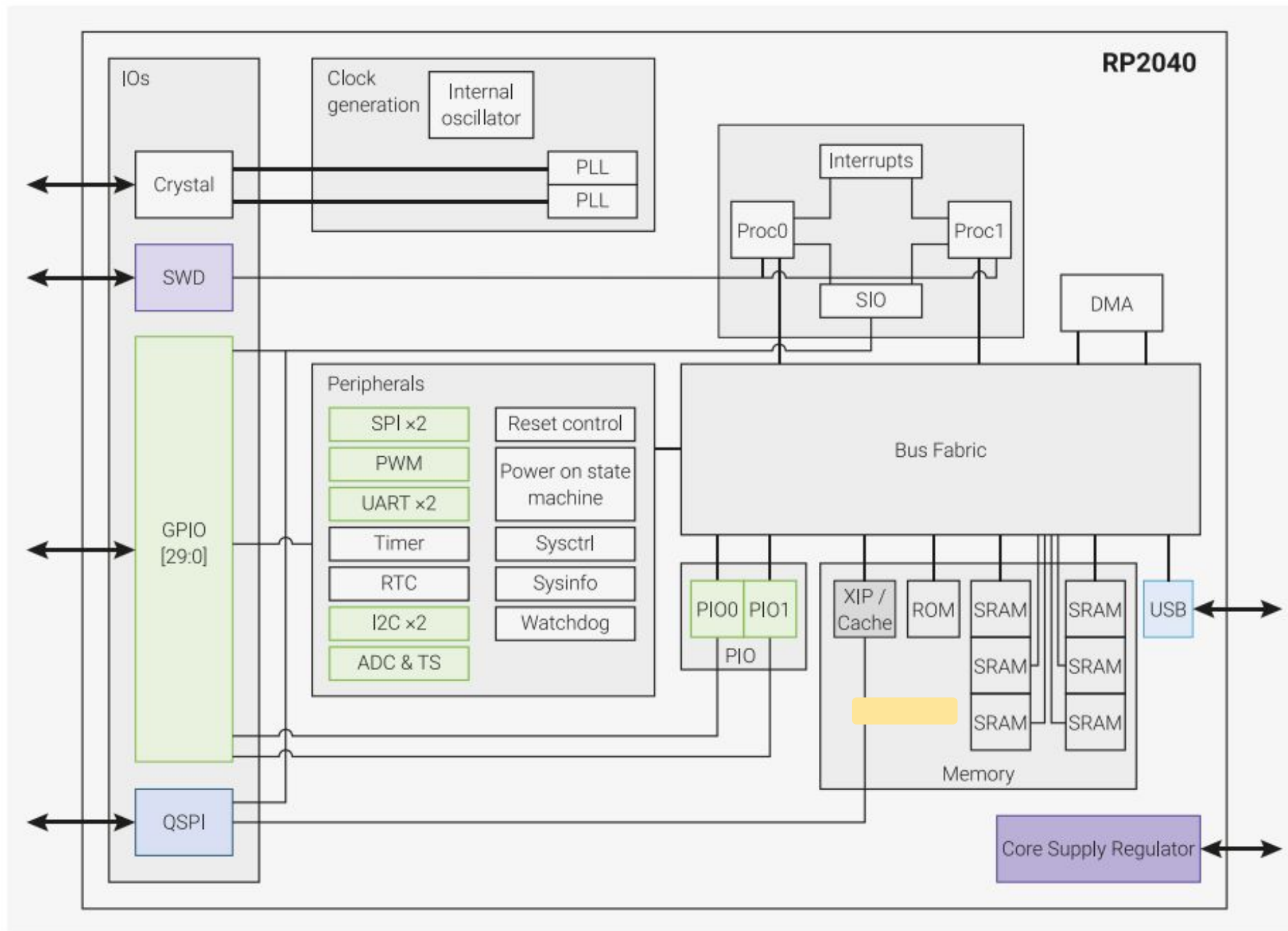
RP2040



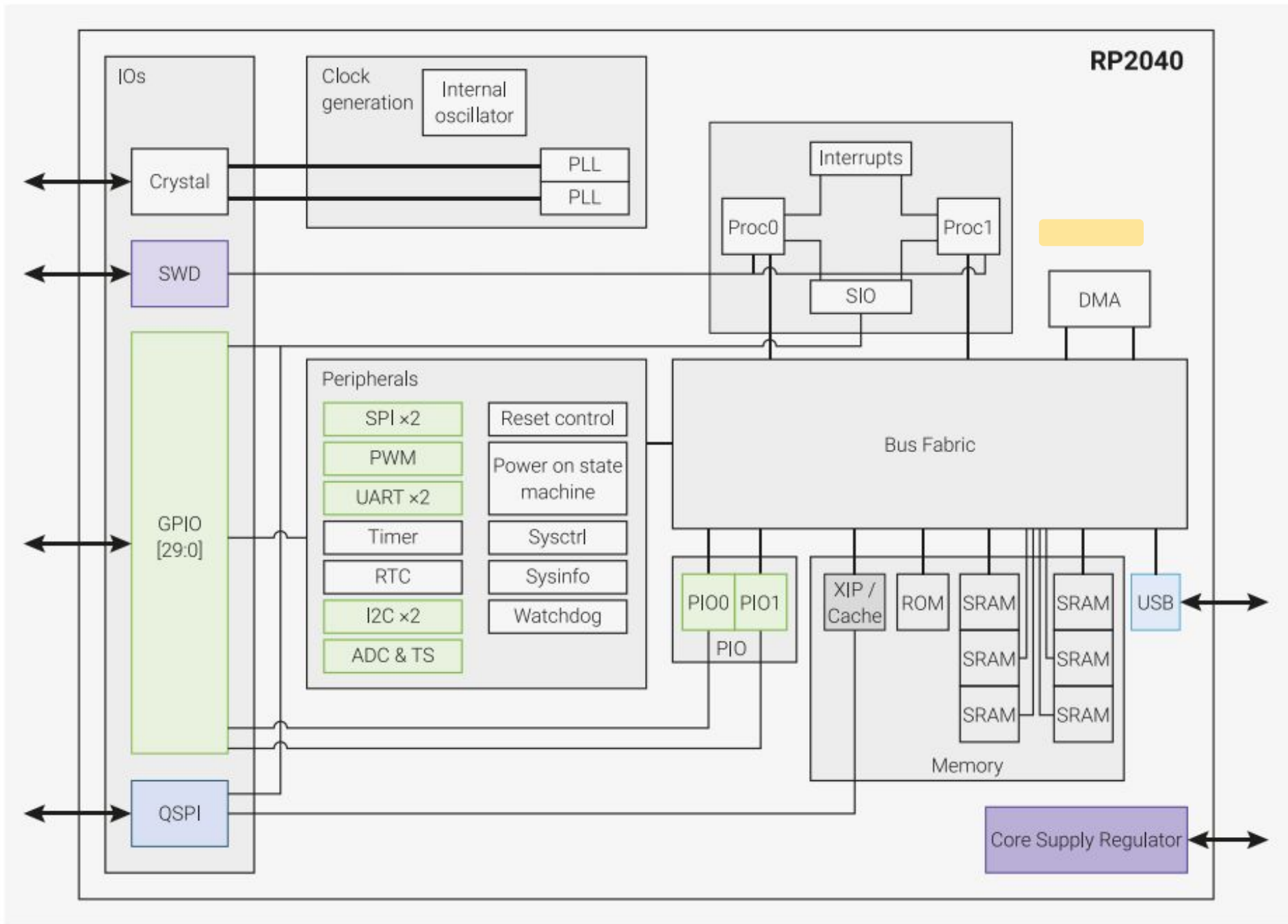
There are 2 processors which can be interrupted and has connection to SIO



Program
mable
input
and
output.

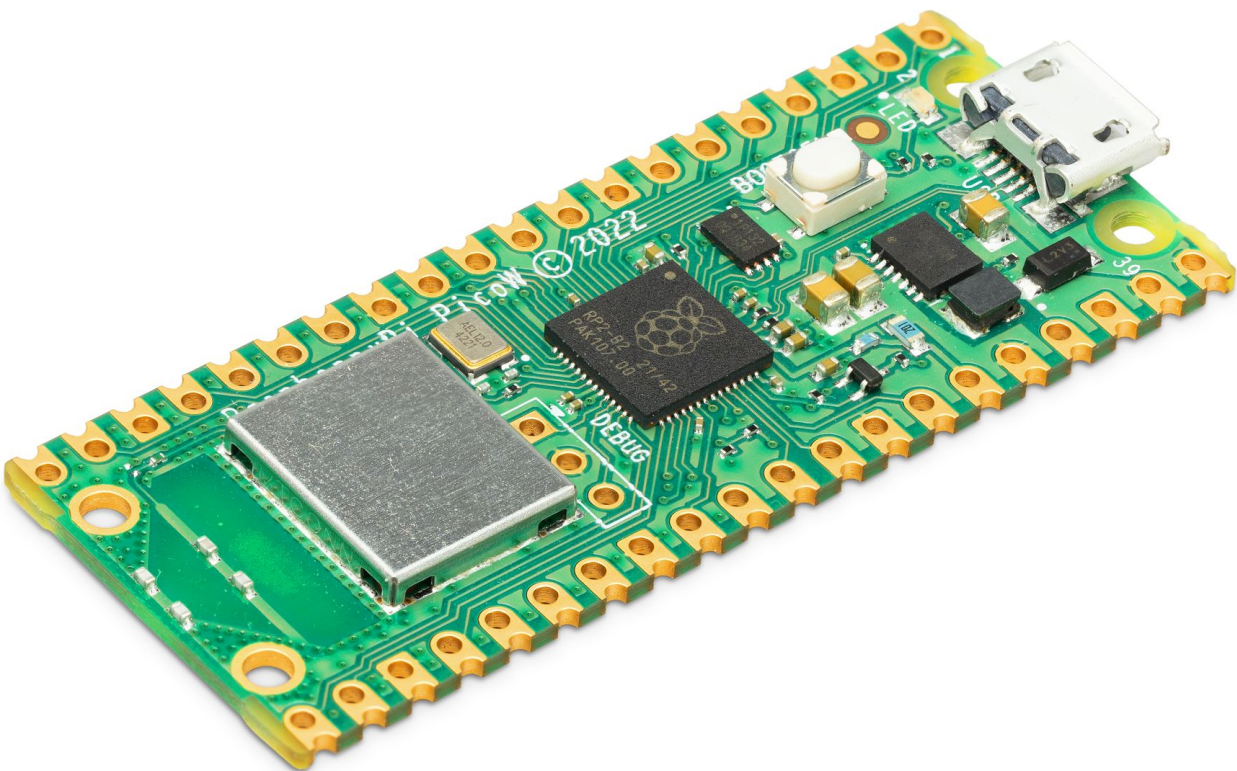


Memories
in RP2040

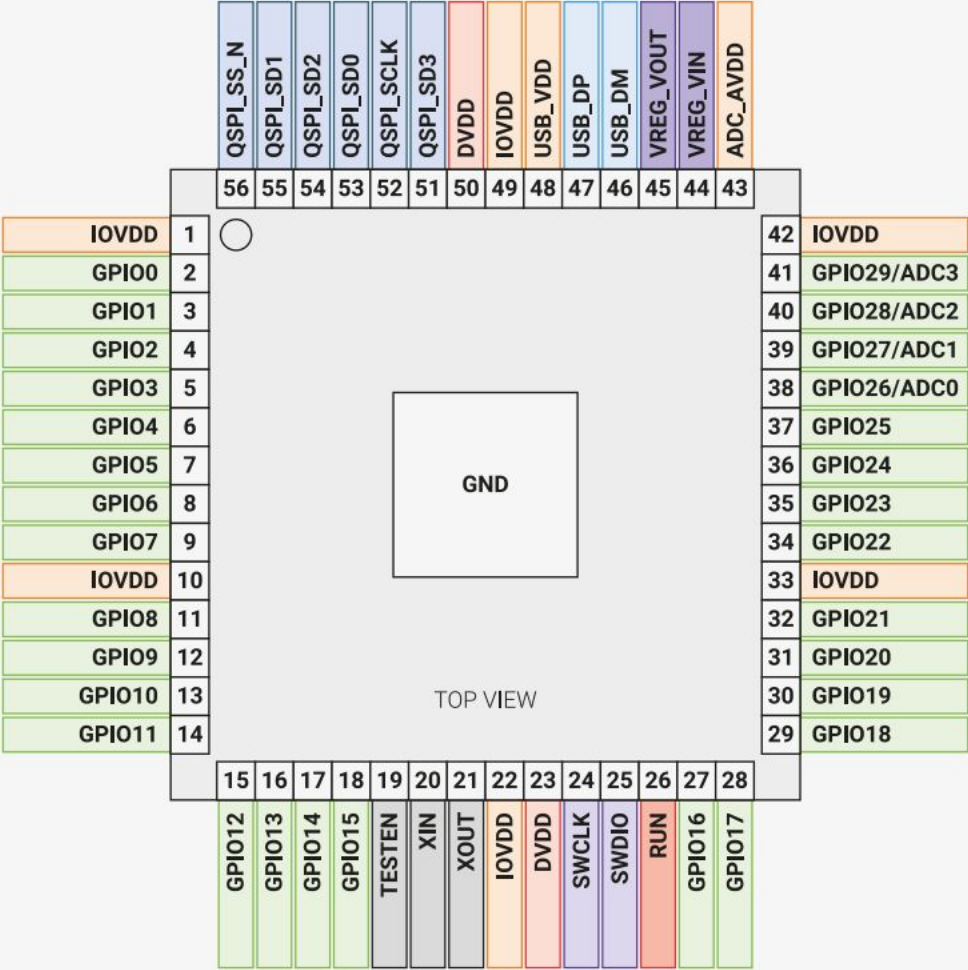


RP2040

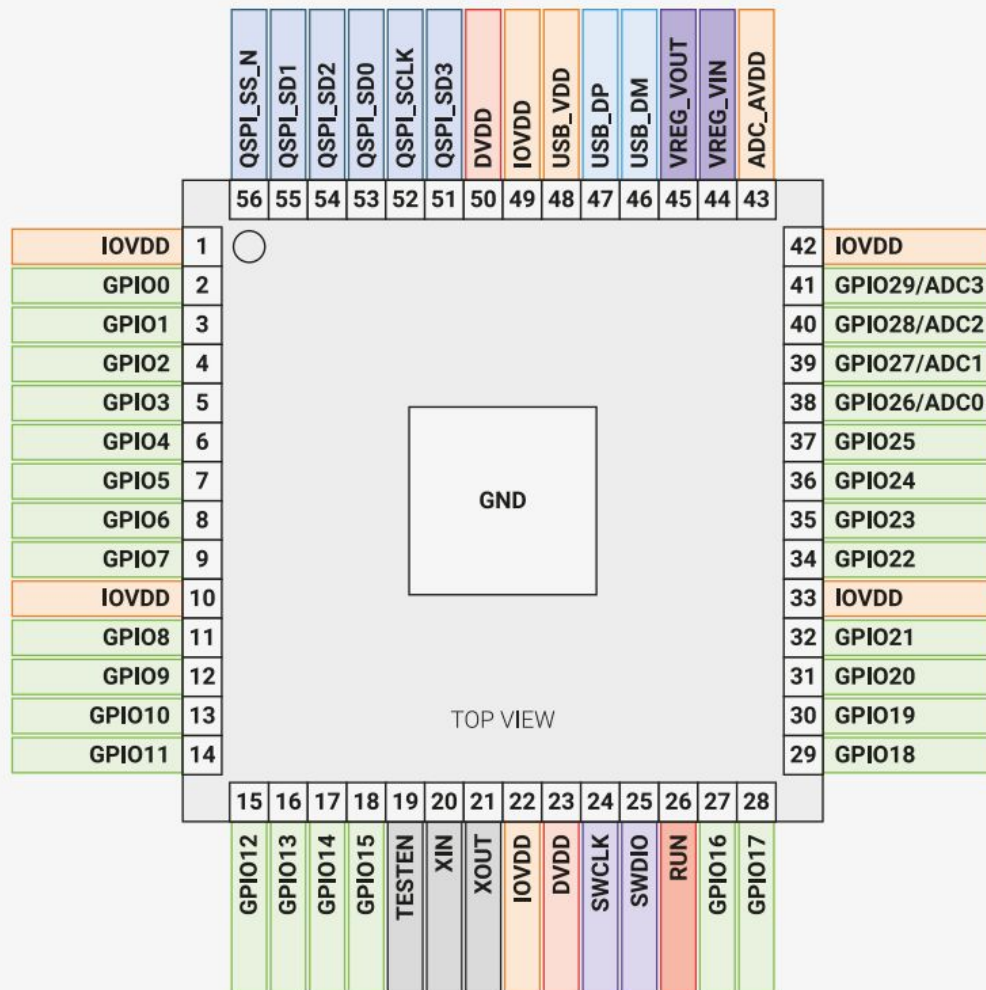
Pin Descriptions



Classify and detail the functions of various pins



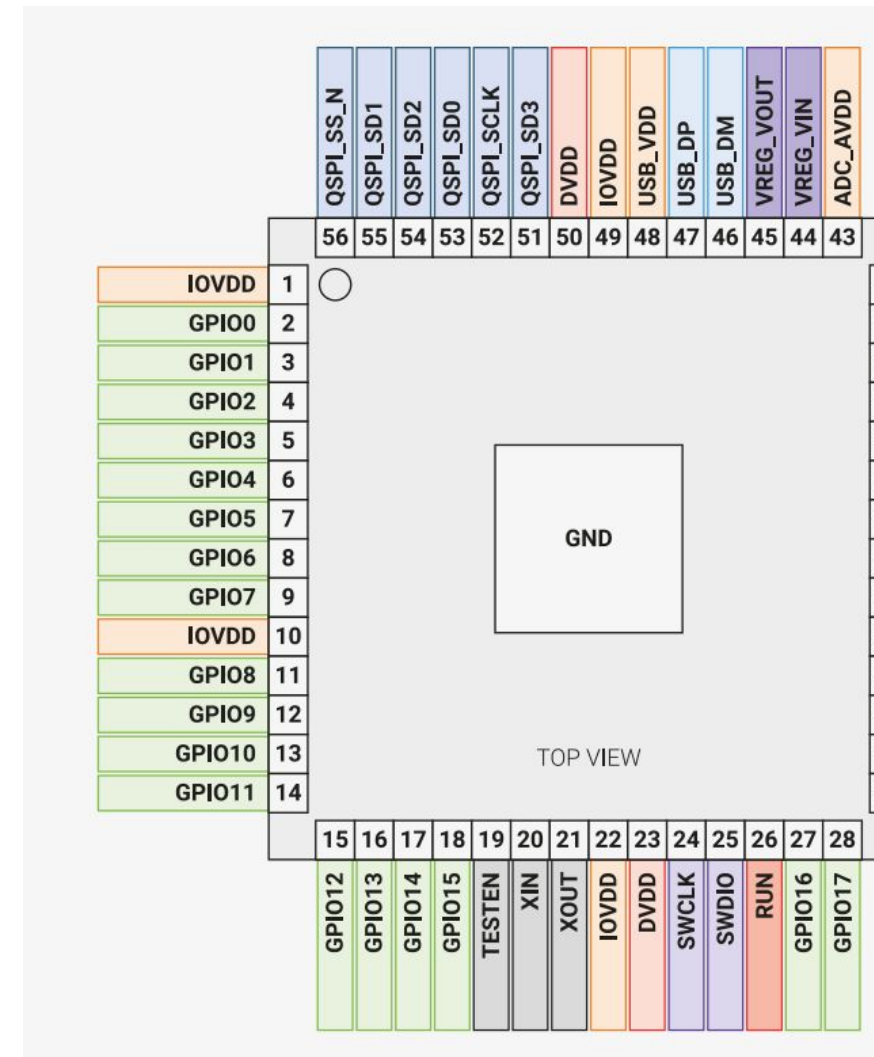
Pin Descriptions



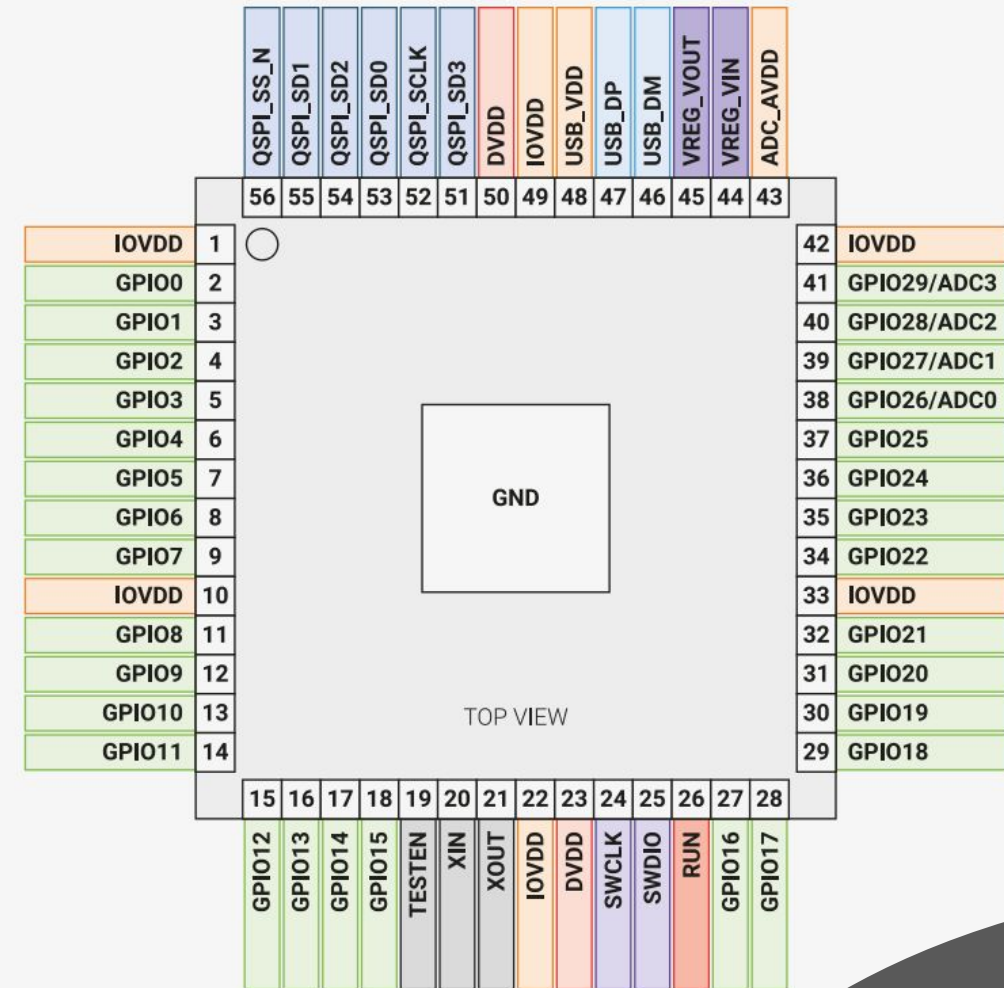
Name	Description
GPIOx	General-purpose digital input and output. RP2040 can connect one of a number of internal peripherals to each GPIO, or control GPIOs directly from software.
GPIOx/ADCy	General-purpose digital input and output, with analogue-to-digital converter function. The RP2040 ADC has an analogue multiplexer which can select any one of these pins, and sample the voltage.
QSPIx	Interface to a SPI, Dual-SPI or Quad-SPI flash device, with execute-in-place support. These pins can also be used as software-controlled GPIOs, if they are not required for flash access.
USB_DM and USB_DP	USB controller, supporting Full Speed device and Full/Low Speed host. A 27Ω series termination resistor is required on each pin, but bus pullups and pulldowns are provided internally.
XIN and XOUT	Connect a crystal to RP2040's crystal oscillator. XIN can also be used as a single-ended CMOS clock input, with XOUT disconnected. The USB bootloader requires a 12MHz crystal or 12MHz clock input.

Pin Descriptions

RUN	Global asynchronous reset pin. Reset when driven low, run when driven high. If no external reset is required, this pin can be tied directly to IOVDD.
SWCLK and SWDIO	Access to the internal Serial Wire Debug multi-drop bus. Provides debug access to both processors, and can be used to download code.
TESTEN	Factory test mode pin. Tie to GND.
GND	Single external ground connection, bonded to a number of internal ground pads on the RP2040 die.
IOVDD	Power supply for digital GPIOs, nominal voltage 1.8V to 3.3V



Name	Description
USB_VDD	Power supply for internal USB Full Speed PHY, nominal voltage 3.3V
ADC_AVDD	Power supply for analogue-to-digital converter, nominal voltage 3.3V
VREG_VIN	Power input for the internal core voltage regulator, nominal voltage 1.8V to 3.3V
VREG_VOUT	Power output for the internal core voltage regulator, nominal voltage 1.1V, 100mA max current
DVDD	Digital core power supply, nominal voltage 1.1V. Can be connected to VREG_VOUT, or to some other board-level power supply.



- VREG_VIN:** Input voltage for the internal voltage regulator circuit.
- IOVDD:** Voltage for the input/output pins used for external interfacing.
- DVDD:** Voltage for the digital core components like the processor and memory.

Pin Descriptions

GPIO Functions

[://www.youtube.com/watch?v=Duel_Oaases&t=745s](https://www.youtube.com/watch?v=Duel_Oaases&t=745s)

To select the functions, watch this video from 44th minute onwards.

<https://docs.micropython.org/en/latest/rp2/quickref.html>

	Function								
GPIO	F1	F2	F3	F4	F5	F6	F7	F8	F9
0	SPI0 RX	UART0 TX	I2C0 SDA	PWM0 A	SIO	PI00	PI01		USB OVCUR DET
1	SPI0 CSn	UART0 RX	I2C0 SCL	PWM0 B	SIO	PI00	PI01		USB VBUS DET
2	SPI0 SCK	UART0 CTS	I2C1 SDA	PWM1 A	SIO	PI00	PI01		USB VBUS EN
3	SPI0 TX	UART0 RTS	I2C1 SCL	PWM1 B	SIO	PI00	PI01		USB OVCUR DET
4	SPI0 RX	UART1 TX	I2C0 SDA	PWM2 A	SIO	PI00	PI01		USB VBUS DET
5	SPI0 CSn	UART1 RX	I2C0 SCL	PWM2 B	SIO	PI00	PI01		USB VBUS EN
6	SPI0 SCK	UART1 CTS	I2C1 SDA	PWM3 A	SIO	PI00	PI01		USB OVCUR DET

<https://wokwi.com/projects/372647173572190209>

GPIO Functions

Each individual GPIO pin can be connected to an internal peripheral via the GPIO functions defined below. Some internal peripheral connections appear in multiple places to allow some system level flexibility. SIO, PIO0 and PIO1 can connect to all GPIO pins and are controlled by software (or software controlled state machines) so can be used to implement many functions. **To select the function use func_sel register, see in page236 of the datasheet**

	Function								
GPIO	F1	F2	F3	F4	F5	F6	F7	F8	F9
0	SPI0 RX	UART0 TX	I2C0 SDA	PWM0 A	SIO	PIO0	PIO1		USB OVCUR DET
1	SPI0 CSn	UART0 RX	I2C0 SCL	PWM0 B	SIO	PIO0	PIO1		USB VBUS DET
2	SPI0 SCK	UART0 CTS	I2C1 SDA	PWM1 A	SIO	PIO0	PIO1		USB VBUS EN
3	SPI0 TX	UART0 RTS	I2C1 SCL	PWM1 B	SIO	PIO0	PIO1		USB OVCUR DET
4	SPI0 RX	UART1 TX	I2C0 SDA	PWM2 A	SIO	PIO0	PIO1		USB VBUS DET
5	SPI0 CSn	UART1 RX	I2C0 SCL	PWM2 B	SIO	PIO0	PIO1		USB VBUS EN
6	SPI0 SCK	UART1 CTS	I2C1 SDA	PWM3 A	SIO	PIO0	PIO1		USB OVCUR DET

```

#define LED 1
unsigned int * gpioOut = (unsigned
int*)0xd0000010;

void setup() {
    // put your setup code here, to run once:
    pinMode(LED, OUTPUT);
    pinMode(3, OUTPUT);
    pinMode(1, OUTPUT);
    pinMode(0, OUTPUT);
    pinMode(6, OUTPUT);
    pinMode(2, OUTPUT);
    pinMode(5, OUTPUT);
    pinMode(4, OUTPUT);
    pinMode(7, OUTPUT);
}

void loop() {
    *gpioOut = 0b11111111;
    delay(1000); // this speeds up the
simulation
    *gpioOut = 0;
    delay(1000);
}

```

```

const int numLEDs = 8; // Number of LEDs
const int ledPins[numLEDs] = {0, 1, 2, 3, 4,
5, 6, 7}; // GPIO pins connected to LEDs

void setup() {
    for (int i = 0; i < numLEDs; i++) {
        pinMode(ledPins[i], OUTPUT); // Set LED
pins as OUTPUT
    }
}

void loop() {
    for (int i = 0; i < numLEDs; i++) {
        digitalWrite(ledPins[i], HIGH); // Turn on
all LEDs simultaneously
    }
    delay(500); // Wait for 500ms

    for (int i = 0; i < numLEDs; i++) {
        digitalWrite(ledPins[i], LOW); // Turn off
all LEDs simultaneously
    }
    delay(500); // Wait for 500ms
}

```

Register based programming to toggle LEDs

<https://wokwi.com/projects/372647917103864833>

<https://docs.micropython.org/en/latest/rp2/quickref.html>

https://www.youtube.com/watch?v=Duel_Oaases&t=745s